

Almost device-independent certification of multipartite quantum states with minimal measurements

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Device-independent certification of quantum states enables the characterization of states within a device under minimal physical assumptions. A major problem in this regard is to certify quantum states using minimal resources. Aiming to address this problem, we consider a multipartite quantum steering scenario involving an arbitrary number of parties, of which only one is trusted, meaning that the measurements performed by this party are known. Consequently, the self-testing scheme is almost device-independent. Importantly, all the parties can only perform two measurements each, which is the minimal number of measurements required to observe any form of quantum nonlocality. Then, we propose steering inequalities that are maximally violated by three major classes of genuinely multipartite entangled (GME) states: graph states of arbitrary local dimension, Schmidt states of arbitrary local dimension, and N -qubit generalized W states. Using the proposed inequalities, we then provide an almost device-independent certification of the above GME states. Restricting to qubits, we also lift our almost device-independent scheme to device-independent self-testing.

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I. INTRODUCTION

Device-independent (DI) quantum information refers to a framework in quantum information theory where protocols and tasks are designed without relying on detailed knowledge or assumptions about the devices used to implement them. In other words, DI protocols aim to achieve certain quantum information processing tasks without specifying the internal workings or characteristics of the quantum devices involved. This approach is particularly valuable for enhancing security and reliability in quantum communication and computation tasks. By removing assumptions about the devices, DI protocols provide a higher level of security against potential attacks and imperfections in the devices themselves. One of the key resources in DI quantum information processing is Bell nonlocality [1,2], which serves as a fundamental ingredient for DI tasks such as quantum cryptography [3], random number generation [4], and state and measurement certification [4–6].

The most complete form of DI certification is referred to as self-testing [7] (see also Ref. [8]). This allows one to infer the structure of the underlying quantum system solely from the observed Bell-type correlations without requiring any knowledge of the measurement implementation. Recently, a great deal of attention has been devoted to finding schemes to self-test entangled quantum states in the bipartite [9–13] and multipartite [14–22] frameworks. In particular, Ref. [20] provides a scheme for self-testing any pure multipartite entangled quantum state and Ref. [21] provides a scheme to self-test any quantum state. Although both results are general, they are based on quantum networks, which significantly impacts their practical applicability.

A major problem in this regard is to find the optimal self-testing scheme in terms of the number of measurements required to certify a particular quantum state from the observed correlations. Clearly, as self-testing is based on quantum nonlocality, the minimal number of measurements per party is two. In the bipartite regime, there are only a few schemes that use two measurements per observer [9,10,12], and even fewer schemes exist in the multipartite scenario [14,18]. Moreover, most of the

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strategies existing in the multipartite scenario are devoted to quantum states that are locally qubits. One of the main challenges lies in designing a suitable Bell inequality that is maximally violated by a given multipartite entangled quantum state. This task becomes even more complex when one aims to construct such an inequality using only two measurements per observer.

To reduce complexity, one can introduce physically motivated assumptions into the considered scenario, leading to the concept of semi-device-independent certification. For example, one may assume that one of the parties is trusted, meaning that their measurement devices are fully characterized. This leads to scenarios typically referred to as one-sided device-independent (1SDI), where the key resource enabling certification is a weaker form of quantum nonlocality known as quantum steering [23–25]. Recent investigations have explored the potential of such scenarios for self-testing quantum states and measurements [26–33]. For this work, we used the multipartite quantum steering framework introduced in Ref. [34].

Here, we address a significantly more challenging problem: certifying genuinely multipartite entangled (GME) states shared among an arbitrary number of parties with arbitrary local dimensions, using only a minimal number of measurements, namely, two per party. To simplify the problem, we make the aforementioned assumption that a single party is trusted, thus making our scheme almost device-independent. Furthermore, we restrict the parties to only choose two measurements each. Under these assumptions, we construct steering inequalities that are maximally violated by three relevant classes of GME states. We then prove that maximal violation of our inequalities can be used for certification of the corresponding states. Since, among the arbitrary number of parties, only one is trusted, the results presented below are almost DI certificates of the quantum state. The first family of states we can certify in this way are graph states of arbitrary local dimension. Graph states form one of the most representative classes of states, which includes, for instance, the well-known Greenberger-Horne-Zeilinger (GHZ) [35], cluster [36], or absolutely maximally entangled (AME) [37] states. Graph states are key resources for many applications, such as measurement-based quantum computing [38,39], quantum metrology [40], or multipartite secret sharing [41,42]. The existing self-testing schemes for graph states are restricted to local dimensions two [14,15] and three [19]. Here, we go beyond these particular dimensions and provide strategies that allow one to certify graph states of arbitrary local dimension, yet at the cost that one of the measurement devices is trusted.

The second class of states we consider here are the Schmidt states of arbitrary local dimension, which also include the aforementioned GHZ states. The Schmidt states are useful for quantum metrology [43] or multipartite secret sharing [41]. A self-testing method for this class

was provided in Ref. [17]. The latter, however, requires more measurements per observer and is thus not optimal as far as physical implementations are concerned. Interestingly, our scheme can also be used to certify two measurements with the untrusted parties to be mutually unbiased using almost no entanglement across multiple parties.

We finally provide a self-testing scheme for the generalized W states (see below for a definition) shared among an arbitrary number of parties, but only with local dimension two. The generalized W states, together with the Schmidt states, belong to a broader class of entangled symmetric multipartite states and find applications, for instance, in quantum metrology [42]. While self-testing methods for the standard N -qubit W state were already introduced in Refs. [16,17], here, by making a mild assumption that only one party is trusted, we can significantly generalize these results to a multiparameter class of genuinely entangled states.

Finally, we show how the above results can be extended to the device-independent regime by taking inspiration from Ref. [20]. More precisely, we consider an extra party, which enables certification of the measurements of the trusted party from violation of the Clauser-Horne-Shimony-Holt (CHSH) Bell inequality.

II. PRELIMINARIES

We consider a 1SDI scenario consisting of N parties, A and $B_1, \dots, B_{N-1}$, which are also called Alice and Bobs, respectively. All the parties are located in space-like separated laboratories and share a quantum state ρ_{AB} , where we denote $\mathbf{B} \equiv B_1 \dots B_{N-1}$. Each party has a measurement device performing measurements on their shares of ρ_{AB} . We additionally assume that Alice's measurement device is trusted, meaning that it performs known measurements; however, we do not assume that her device performs full tomography over her share of ρ_{AB} . In contrast, the measurement devices belonging to the $N - 1$ Bobs are untrusted and are treated as “black boxes.” The scenario we just described is also called *local steering* [44], in which the untrusted parties perform local measurements in their respective states to steer the state owned by the trusted party.

In this work, we consider that the trusted party performs the measurements given by the d -dimensional generalization of the Pauli Z and X observables,

$$Z_d = \sum_{i=0}^{d-1} \omega^i |i\rangle\langle i|, \quad X_d = \sum_{i=0}^{d-1} |i+1\rangle\langle i|, \quad (1)$$

which, for ease of notation, will be denoted as Z and X , respectively. Figure 1 illustrates the scenario considered.

We denote the inputs and outputs of the parties A and $B_1, \dots, B_{N-1}$ with x and $y_1, \dots, y_{N-1}$ and a and

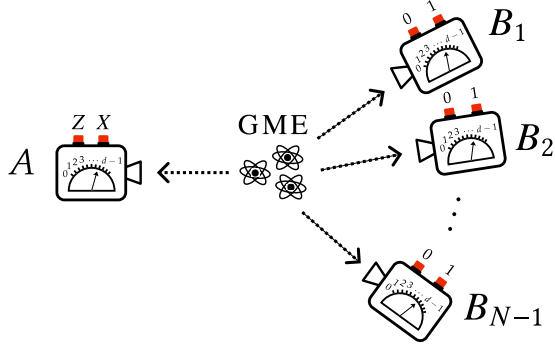


FIG. 1. Depiction of the local steering scenario that we consider. A GME state is shared among N parties: one Alice and $N - 1$ Bobs. While Alice performs measurements Z and X , all the Bobs perform unknown measurements.

$b_1, \dots, b_{N-1}$, respectively. For convenience, we use the notation $\mathbf{y} \equiv y_1, \dots, y_{N-1}$ and $\mathbf{b} \equiv b_1, \dots, b_{N-1}$. The correlations observed by the parties when they repeat their measurements many times are captured by the set of probability distributions

$$\vec{p} = \{p(a, \mathbf{b}|x, \mathbf{y})\}, \quad (2)$$

where $p(a, \mathbf{b}|x, \mathbf{y})$ represents the probability of obtaining results a and $\mathbf{b}$ given that the measurements x and $\mathbf{y}$ have been carried out. According to Born's rule, the latter can be expressed as

$$p(a, \mathbf{b}|x, \mathbf{y}) = \text{Tr} \left[\rho_{AB} \left(M_{a|x} \otimes \bigotimes_{i=1}^{N-1} N_{b_i|y_i}^{(i)} \right) \right], \quad (3)$$

where $M_{a|x}$ and $N_{b_i|y_i}^{(i)}$ are the measurement operators representing the measurements of Alice and all Bobs B_i , respectively; recall that they are positive semidefinite and sum to identity on their corresponding Hilbert spaces.

In our multipartite 1SDI scenario, the presence of local quantum steering is demonstrated if the probability distribution $p(a, \mathbf{b}|x, \mathbf{y})$ cannot be described by a local hidden state (LHS) model [23,44], i.e., if it cannot be written as

$$p(a, \mathbf{b}|x, \mathbf{y}) = \sum_{\lambda} p(\lambda) \text{Tr} [M_{a|x} \rho_A^{\lambda}] p(\mathbf{b}|\mathbf{y}, \lambda), \quad (4)$$

where λ is a hidden variable that follows the probability distribution $p(\lambda)$. The hidden variable λ determines the state ρ_A^{λ} received by Alice and also the responses $p(\mathbf{b}|\mathbf{y}, \lambda)$ obtained by $N - 1$ Bobs. If the set of probability distributions $\{p(a, \mathbf{b}|x, \mathbf{y})\}$ cannot be described by the above model, this leads to violation of the linear steering inequality $\mathcal{B}[\{p(a, \mathbf{b}|x, \mathbf{y})\}] \leq \beta_L$, where $\mathcal{B}$ is a linear combination of $\{p(a, \mathbf{b}|x, \mathbf{y})\}$ and β_L is the LHS bound, also called the classical bound [24], which is the maximal value of the functional $\mathcal{B}$ over the LHS models (4).

In this work, we express steering inequalities in terms of generalized expectation values, which are defined through $p(a, \mathbf{b}|x, \mathbf{y})$ in the following way:

$$\langle A_{k|x} \mathbf{B}_{l|\mathbf{y}} \rangle = \sum_{a, \mathbf{b}=0}^{d-1} \omega^{ak+b_1l_1+\dots+b_{N-1}l_{N-1}} p(a, \mathbf{b}|x, \mathbf{y}), \quad (5)$$

where $\mathbf{B}_{l|\mathbf{y}} \equiv (B_1)_{l_1|y_1} \dots (B_{N-1})_{l_{N-1}|y_{N-1}}$ with $k, l_1, \dots, l_{N-1} \in \{0, \dots, d-1\}$, and $\omega = \exp(2\pi i/d)$. If the probabilities $p(a, \mathbf{b}|x, \mathbf{y})$ are expressed through Eq. (3), we can write the expectation values as

$$\langle A_{k|x} \mathbf{B}_{l|\mathbf{y}} \rangle = \text{Tr} \left[\rho_{AB} A_x^k \otimes \bigotimes_{i=1}^{N-1} (B_i)_{l_i|y_i} \right], \quad (6)$$

where A_x^k is the k th power of the unitary observable A_x , which, as a reminder, we assume to be the generalized Pauli observables Z and X . Then, $(B_i)_{l_i|y_i}$ are operators representing the measurements of B_i , which are defined through the Fourier transforms of the measurement operators $N_{b_i|y_i}^{(i)}$, i.e.,

$$(B_i)_{l_i|y_i} = \sum_{b_i} \omega^{b_i l_i} N_{b_i|y_i}^{(i)}. \quad (7)$$

In fact, since we can write $N_{b_i|y_i}^{(i)}$ as an inverse Fourier transform of $(B_i)_{l_i|y_i}$, we can also refer to $\{(B_i)_{l_i|y_i}\}_{l_i=0}^{d-1}$ as the measurements of the i th Bob. Moreover, if $N_{y_i}^{(i)}$ are projective, then $(B_i)_{l_i|y_i}$ are the l_i th power of a unitary observable $B_{i,y_i} := (B_i)_{1|y_i}$; that is, $(B_i)_{l_i|y_i} = (B_{i,y_i})^{l_i}$ [13].

In what follows, we present steering inequalities that are maximally violated by three classes of GME states: the N -qudit graph states, the N -qudit Schmidt states, and the generalized N -qubit W states. Recall that a GME state is one that cannot be written as a tensor product of two other pure states corresponding to an arbitrary bipartition of all the parties into two disjoint and nonempty subsets. More precisely, an N -partite state $|\psi_{AB}\rangle$ is GME if $|\psi_{AB}\rangle \neq |\alpha\rangle_Q \otimes |\beta\rangle_{\bar{Q}}$ for any two states $|\alpha\rangle_Q$ and $|\beta\rangle_{\bar{Q}}$ and any bipartition of the parties A and $B_1, \dots, B_{N-1}$ into two nonempty, disjoint sets Q and $\bar{Q}$.

Maximal violations of steering inequalities play a critical role in the self-testing results presented below. When the observed correlations $\vec{p}$ exhibit a maximal violation, they allow us to infer both the observer's state and the performed measurements up to local isometries. With that in mind, let us formally define multipartite 1SDI self-testing in a manner similar to that introduced in Ref. [30]. We assume here that the measurements of the untrusted parties are projective based on dilation arguments. At the same time, we do not assume the shared state ρ_{AB} to be purifiable by the untrusted Bobs as, due to the Schmidt

decomposition between A and B , this would imply that the dimension of B 's joint Hilbert space is the same as that of Alice, i.e., d . Nonetheless, we can always consider a purification $|\psi\rangle_{ABE}$ with an extra system E such that $\rho_{AB} = \text{Tr}_E(|\psi\rangle\langle\psi|_{ABE})$, which is not possessed by any of the parties. Alice's measurements are projective and fixed as A_x , while the measurements of the Bobs are arbitrary and represented by the operators $(B_i)_{l_i|v_i}$. The goal is to certify that the state shared between Alice and Bob and the measurements performed by the Bobs are equivalent, up to certain equivalences, to a reference state $|\psi'\rangle_{AB'} \in (\mathbb{C}^d)^{\otimes N}$ and to the reference measurements represented by $(B'_i)_{l_i|v_i}$ defined on $\mathbb{C}^d$, respectively, which all give rise to the same correlations $\vec{p}$. More formally, we say that $|\psi'\rangle_{AB'}$ and $(B'_i)_{l_i|v_i}$ are certified from the observed $\vec{p}$ if there exists a unitary $U_i : \mathcal{H}_{B_i} \rightarrow \mathcal{H}_{B'_i}$ for each Bob B_i such that

$$\left(\mathbb{1}_A \otimes \bigotimes_{i=1}^{N-1} U_i \right) |\psi\rangle_{ABE} = |\psi'\rangle_{AB'} \otimes |\phi\rangle_{B''E} \quad (8)$$

and

$$U_i (B_i)_{l_i|v_i} U_i^\dagger = (B'_i)_{l_i|v_i} \otimes \mathbb{1}_{B''}, \quad \forall i, \quad (9)$$

where Bob's Hilbert spaces decompose into tensor products $\mathcal{H}_{B_i} = (\mathbb{C}^d)_{B'_i} \otimes \mathcal{H}_{B''_i}$, in which $\mathcal{H}_{B'_i}$ is an auxiliary Hilbert space of unknown dimension and $|\phi\rangle_{B''E}$ is a junk state from $\mathcal{H}_{B''} \otimes \mathcal{H}_E$. Let us remark here that the measurements can only be certified on the local supports of the state; thus, without loss of generality, we assume that the local density matrices of the joint state $|\psi\rangle_{ABE}$ are full rank.

III. 1SDI CERTIFICATION OF MULTIPARTITE QUANTUM STATES

A. Graph states

Let us begin by defining graph states of arbitrary local dimension [45–47]. Consider a multigraph $G = (V, E, \Gamma)$ (see Fig. 2), where V is a set of all the vertices of the graph, whereas E is a set of pairs of vertices that are connected by edges. Finally, Γ is a matrix with elements $\gamma_{i,j} \in \{0, \dots, d-1\}$ specifying the multiplicities of the edges between vertices i and j ; notice that $\gamma_{i,j} = \gamma_{j,i}$ for any pair i and j , and that $\gamma_{i,i} = 0$. For our purpose, we consider graphs that are connected, that is, those in which there exists a path between any pair of vertices. Then, the graph state $|G\rangle \in (\mathbb{C}^d)^{\otimes N}$ corresponding to the multigraph $G = (V, E, \Gamma)$, where N is the number of vertices, is constructed as follows:

(a) Each vertex of G is associated with a single qudit state $|+_i^d\rangle = (1/\sqrt{d}) \sum_{i=0}^{d-1} |i\rangle$, where $\{|i\rangle\}$ is the

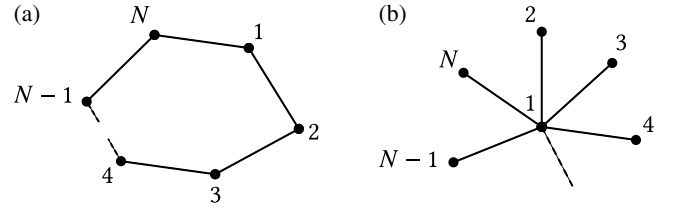


FIG. 2. Examples of graphs: (a) ring graph, (b) star graph corresponding to the N -qudit GHZ state up to local unitaries.

d -dimensional computational basis. The total state of N qudits is thus

$$|+_V^d\rangle = \bigotimes_{i=1}^N |+_i^d\rangle. \quad (10)$$

(b) For each edge $e = \{i, j\}$ connecting the vertices i and j of multiplicity $\gamma_{i,j}$, one applies the controlled unitary

$$Z_{i,j}^{\gamma_{i,j}} = \sum_{q=0}^{d-1} |q\rangle\langle q|_i \otimes \left(Z_j^{\gamma_{i,j}} \right)^q \quad (11)$$

to the state $|+_V^d\rangle$.

Consequently, a graph state corresponding to the multigraph $G = (V, E)$ is defined as

$$|G\rangle = \prod_{\{i,j\} \in E} Z_{i,j}^{\gamma_{i,j}} |+_V^d\rangle. \quad (12)$$

An equivalent way to represent graph states is by using the stabilizer formalism [48]. Let us consider a multigraph $G = (V, E)$ and associate the following operator with each vertex i :

$$\bar{S}(G)_{d,i} = X_i \otimes \bigotimes_{j \in n(i)} Z_j^{\gamma_{i,j}} \quad (i = 1, 2, \dots, N), \quad (13)$$

where $n(i)$ is the set of vertices connected to the vertex i , which is also called the neighborhood of i . The graph state $|G\rangle$ is the unique eigenstate of all the N stabilizing operators $S_i(G)$ corresponding to the eigenvalue 1. For example, the GHZ states are graph states.

Before presenting our first results, we note that fully device-independent schemes for graph states of local dimension two or three have already been proposed in Refs. [14,15] and [19], respectively, whereas no scheme is known beyond these specific dimensions. Here, we provide strategies that allow for the certification of graph states of arbitrary local dimension, albeit under the assumption that one of the measurement devices is trusted.

Let us now move on to constructing steering inequalities that are maximally violated by graph states and then use

these inequalities for self-testing. Let us consider a graph state $|G\rangle$ associated with a graph G . We construct the steering operator from the corresponding stabilizing operators (13). We then assume the first vertex to be trusted and label it as 0; thus, the measurement operators acting on the first site are known to be Z_d and X_d (1). Next, we replace Z_d and X_d on the j th vertex with arbitrary d -outcome observables $B_{j,0}$ and $B_{j,1}$, respectively. The steering operator is then given by

$$\begin{aligned} \mathcal{I}_d(G, N) = & X_A \otimes \bigotimes_{j \in n(0)} (B_{j,0})^{\gamma_{0,j}} \\ & + \sum_{j \in n(0)} Z_A^{\gamma_{j,0}} \otimes B_{j,1} \otimes \bigotimes_{j' \in n(j) \setminus \{0\}} (B_{j',0})^{\gamma_{j,j'}} \\ & + \sum_{j \notin n(0) \cup \{0\}} B_{j,1} \otimes \bigotimes_{j' \in n(j)} (B_{j',0})^{\gamma_{j,j'}} + \text{H.c.}, \quad (14) \end{aligned}$$

where H.c. denotes the Hermitian conjugate of the operator on its left-hand side. Notice that for a given graph $G = (V, E)$, the first term of the above operator corresponds to the stabilizer when choosing vertex 0, the second term consists of the stabilizers for vertices connected to the 0th vertex, and the third term consists of the remaining ones. Also notice that in the above operator, $\gamma_{i,j} = \gamma_{j,i}$ for any i and j . Thus, the steering inequality is given by

$$\langle \mathcal{I}_d(G, N) \rangle \leq \beta_L, \quad (15)$$

where β_L is the maximum value attainable using the LHS model. In Fact 1 of Appendix A, we provide an upper bound to β_L in terms of a simple optimization problem that can be numerically evaluated for any graph G and any number of parties N and outcomes d .

The quantum bound β_Q of the steering functional $\langle \mathcal{I}_d(G, N) \rangle$ is equal to the number of terms it consists of, which is twice the number of the corresponding stabilizing operators, that is, $\beta_Q = 2N$. Clearly, the quantum bound is attained by the graph state $|G\rangle$ and observables $B_{j,0} = Z$ and $B_{j,1} = X$ for any j . Importantly, it follows from Fact 1 of Appendix A that $\beta_L < \beta_Q$, and, consequently, the proposed steering inequality is nontrivial for any graph state.

Now, if all the parties observe that $\langle \mathcal{I}_d(G, N) \rangle = 2N$, then we can establish the following theorem.

Theorem 1. Assume that for a given graph G , the corresponding functional (14) reaches its maximal value $\langle \mathcal{I}_d(G, N) \rangle = 2N$ for a state $|\psi_{ABE}\rangle$ and observables B_{i,y_i} . Then, the following statement holds true for any d : $\mathcal{H}_{B_i} = (\mathbb{C}^d)_{B'_i} \otimes \mathcal{H}_{B''_i}$, where $\mathcal{H}_{B''_i}$ is some finite-dimensional

Hilbert space, and there exists a local unitary transformation on Bob's side $U_i : \mathcal{H}_{B_i} \rightarrow \mathcal{H}_{B_i}$, such that for any i ,

$$U_i B_{i,0} U_i^\dagger = Z \otimes \mathbb{1}_{B''_i}, \quad U_i B_{i,1} U_i^\dagger = X \otimes \mathbb{1}_{B''_i}, \quad (16)$$

where B'' denotes Bob's auxiliary system and the state $|\psi\rangle_{ABE}$ is given by

$$\left(\mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_i \right) |\psi\rangle_{ABE} = |G\rangle_{AB'} \otimes |\xi\rangle_{B''E}, \quad (17)$$

where $|\xi\rangle_{B''E}$ denotes the auxiliary state corresponding to subsystems B'' and E .

The proof of the above theorem is given in Appendix A.

In practical scenarios, it is important to investigate the robustness of self-testing statements, such as those introduced in our work, against experimental imperfections, including noise in the preparation device and imperfect measurements; however, deriving analytical robustness bounds is typically a highly demanding task, especially for systems of arbitrary local dimension. In the simplest case of graph states of local dimension two, we can instead exploit Jordan's lemma, which significantly simplifies the problem and allows us to obtain the following robustness bound.

Theorem 2. Assume that the steering inequality (A3) attains a value close to the maximal violation, $\langle \mathcal{I}_2(G, N) \rangle \geq \beta_Q - \varepsilon$, when the 0th party Alice measures the observables $A_0 = Z$ and $A_1 = X$. Then, there exist local unitary operations U_i such that the following inequality is satisfied:

$$\| |\psi'\rangle_{ABE} - |G\rangle_{AB'} \otimes |\xi\rangle_{B''E} \| \leq (\sqrt{N} + g/2)\sqrt{\varepsilon}, \quad (18)$$

where, as before,

$$|\psi'\rangle_{ABE} = \mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_i |\psi\rangle_{ABE}, \quad (19)$$

$|G\rangle_{AB'}$ is the graph state to be certified, and $|\xi\rangle_{B''E}$ is the auxiliary state. Finally, g is a factor depending on the particular graph (c.f. Theorem A2 in Appendix A).

A detailed version of the theorem and the proof is given in Appendix A.

B. Schmidt states

Every pure bipartite state $|\psi\rangle_{AB}$ admits a Schmidt decomposition; that is, there exist orthonormal bases $|i\rangle_A$ and $|i\rangle_B$ in $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively, such that $|\psi\rangle_{AB} =$

$\sum_i \alpha_i |i\rangle_A \otimes |i\rangle_B$, where $\alpha_i > 0$ and $\sum_i \alpha_i^2 = 1$. An analogous decomposition in the multipartite setting is in general impossible. The multipartite pure states that can be represented via Schmidt decomposition are referred to as Schmidt states [17]. Under the action of local unitary operators, an N -qudit Schmidt state $|\psi(\alpha)\rangle_{AB}$ can be brought to the following form:

$$|\psi(\alpha)\rangle_{AB} = \sum_{i=0}^{d-1} \alpha_i |i\rangle^{\otimes N}, \quad (20)$$

with $\alpha \equiv \alpha_1, \dots, \alpha_N$ such that $0 < \alpha_i < 1$ for all i and $\sum_{i=0}^{d-1} \alpha_i^2 = 1$. Because $\alpha_i > 0$ for all i , the Schmidt rank of the state is maximal, which ensures that it is a truly N -qudit state.

Importantly, a self-testing scheme for multipartite Schmidt states in the device-independent regime is already known and was proposed in Ref. [17]. That approach, however, requires more than two measurements per observer and is therefore suboptimal from an implementation perspective. Specifically, $N - 1$ parties perform three measurements each, while the remaining party performs four. In this work, we exploit the assumption that one of the observers is trusted to reduce the number of measurements per observer to two.

Inspired by Ref. [31], let us now introduce a general class of steering inequalities that are maximally violated by the above Schmidt state for a given α . For this purpose, we consider the following steering operator:

$$\begin{aligned} \mathcal{S}_d(\alpha, N) = \sum_{k=1}^{d-1} \left[\left(\sum_{i=1}^{N-1} A_0^k \otimes B_{i,0}^k \right) \right. \\ \left. + \gamma(\alpha) A_1^k \otimes \bigotimes_{i=1}^{N-1} B_{i,1}^k + \delta_k(\alpha) A_0^k \right], \end{aligned} \quad (21)$$

where the coefficients $\gamma(\alpha)$ and $\delta_k(\alpha)$ are given by

$$\gamma(\alpha) = d \left[\sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j} \right]^{-1}, \quad (22)$$

$$\delta_k(\alpha) = -\frac{\gamma(\alpha)}{d} \sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j} \omega^{k(d-j)}. \quad (23)$$

In our scenario, we consider that all Bobs perform two arbitrary measurements, denoted by $B_{i,0}$ and $B_{i,1}$, whereas Alice's measurement device is trusted and measures the generalized Pauli observables $A_0 = Z$ and $A_1 = X$.

The corresponding steering inequality is given by

$$\langle \mathcal{S}_d(\alpha, N) \rangle \leq \beta_L. \quad (24)$$

Although it is a difficult task to analytically compute the maximal classical value of the expression β_L , in Fact 3 of Appendix B, we provide an upper bound on β_L in terms of an optimization problem that can be numerically evaluated. Then, the quantum bound β_Q of $\langle \mathcal{S}_d(\alpha, N) \rangle$ is given by $\beta_Q = (d-1)(N-1) + 1$ and can be attained by $|\psi(\alpha)\rangle$ with observables $B_{j,0} = Z_d$ and $B_{j,1} = X_d$ for any j . Importantly, from Fact 3, we can also conclude that $\beta_L < \beta_Q$ and thus the proposed steering inequality is nontrivial.

Let us now state our result.

Theorem 3. Assume that the steering inequality (24) is maximally violated, that is, $\langle \mathcal{S}_d(\alpha, N) \rangle = (d-1)(N-1) + 1$. Then, the following statement holds true for any d : $\mathcal{H}_{B_i} = (\mathbb{C}^d)_{B'_i} \otimes \mathcal{H}_{B''_i}$, where $\mathcal{H}_{B'_i}$ is a finite-dimensional Hilbert space, and there exists a local unitary transformation on Bob's side $U_i : \mathcal{H}_{B_i} \rightarrow \mathbb{C}^d \otimes \mathcal{H}_{B''_i}$ such that for any i ,

$$U_i B_{i,0} U_i^\dagger = Z \otimes \mathbb{1}_{B''_i}, \quad U_i B_{i,1} U_i^\dagger = X \otimes \mathbb{1}_{B''_i}, \quad (25)$$

where B''_i denotes Bob's auxiliary system, and the state $|\psi\rangle_{AB}$ is given by

$$\left(\mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_i \right) |\psi_N\rangle = |\psi(\alpha)\rangle_{AB'} \otimes |\xi\rangle_{B''E}, \quad (26)$$

where $|\xi\rangle_{B''E}$ denotes the auxiliary state.

The proof of the above theorem can be found in Appendix B.

C. N -qubit generalized W states

The third and last class of states that we consider comprises the N -qubit generalized W states. As the name suggests, they represent a generalization of the well-known W state $|\psi_W\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$ and have the following form:

$$|\psi_W(\alpha)\rangle = \sum_{l=0}^{N-1} \alpha_{l+1} |0 \dots, 0, 1_l, 0, \dots, 0\rangle, \quad (27)$$

such that $\alpha_i > 0$ for every i and $\sum_{i=1}^N \alpha_i^2 = 1$. We introduce a steering inequality that is maximally violated by

the above state through the following steering operator:

$$\begin{aligned} \mathcal{W}_N(\alpha) = & -2Z_A \otimes \bigotimes_{k=1}^{N-1} B_{k,0} + \sum_{l=1}^{N-1} [Z_A \otimes \mathbb{1}_l \otimes (\mathbb{1} - P_l) \\ & + (\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l) \otimes P_l] \\ & + \sum_{l=1}^{N-1} [\mathbb{1}_A \otimes B_{l,0} \otimes (\mathbb{1} - P_l) + \mathbb{1}_A \otimes \mathbb{1}_l \otimes P_l], \end{aligned} \quad (28)$$

where

$$\gamma_l = \frac{2\alpha_{l+1}\alpha_1}{\alpha_{l+1}^2 + \alpha_1^2}, \quad \delta_l = \frac{\alpha_{l+1}^2 - \alpha_1^2}{\alpha_{l+1}^2 + \alpha_1^2}, \quad (29)$$

for every $l \in \{1, \dots, N-1\}$,

$$P_l = \frac{1}{2^{N-2}} \bigotimes_{k=1, k \neq l}^{N-1} (\mathbb{1}_k + B_{k,0}), \quad (30)$$

and $B_{k,0}$ and $B_{k,1}$ are the measurements acting on the k th qubit. Similarly, $\mathbb{1}_l$ is the identity operator acting on the l th qubit. The corresponding steering inequality is given by

$$\langle \mathcal{W}_N(\alpha) \rangle \leq \beta_L. \quad (31)$$

While, as before, we are unable to compute the LHS bound β_L , we can prove the inequality to be nontrivial by demonstrating that observation of the maximal quantum value of $\langle \mathcal{W}_N(\alpha) \rangle$, which is $\beta_Q = 2N$ (see Appendix C for a proof), allows one to self-test $|\psi_W(\alpha)\rangle$. In fact, we can prove the following statement.

Theorem 4. Assume that $\langle \mathcal{W}_N(\alpha) \rangle = 2N$ for a state $|\psi\rangle_{ABE}$ and Bob's observables B_{i,y_i} . Then, $\mathcal{H}_{B_i} = \mathbb{C}^2 \otimes \mathcal{H}_{B''_i}$, where $\mathcal{H}_{B''_i}$ is some finite-dimensional Hilbert space, and there exists a local unitary transformation on Bob's side $U_i : \mathcal{H}_{B_i} \rightarrow \mathbb{C}^2 \otimes \mathcal{H}_{B''_i}$, such that

$$U_i B_{i,0} U_i^\dagger = Z \otimes \mathbb{1}_{B''_i}, \quad U_i B_{i,1} U_i^\dagger = X \otimes \mathbb{1}_{B''_i} \quad (32)$$

for every i , where B'' denotes Bob's auxiliary system and the state $|\psi\rangle_{ABE}$ is given by

$$\left(\mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_i \right) |\psi\rangle_{ABE} = |\psi_W(\alpha)\rangle_{AB'} \otimes |\xi\rangle_{B''E}, \quad (33)$$

where $|\xi\rangle_{B''E}$ denotes the auxiliary state.

As before, the proof of this statement is deferred to Appendix C.

Let us notice that self-testing methods for the standard N -qubit W state (i.e., one for which $\alpha_l = 1/\sqrt{N}$ for every $l = 1, \dots, N$) were introduced in Refs. [16,17]. Here, by making a mild assumption that a single party is trusted, we can significantly generalize these results to a multiparameter class of genuinely entangled states.

IV. EXTENDING TO DEVICE INDEPENDENCE

We now show how our 1SDI certification methods for multipartite quantum states can be extended to the device-independent regime, restricting, for simplicity, to the case where the quantum states are locally qubits. To this end, we combine our quantum steering scenario with the network-assisted approach presented in Ref. [20]. More precisely, we introduce an additional party, Charlie, between Alice and the Bobs, as well as an extra source of quantum particles distributing a state $|\phi\rangle_{AC}$ to Alice and Charlie (cf. Fig. 1). The aim of Charlie is to play the CHSH game with Alice to certify her measurements and also to teleport the local state $\rho_A = \text{Tr}_{\mathbf{B}} \rho_{AB}$ of the multipartite state ρ_{AB} to Alice. Charlie performs three measurements (cf. Fig. 3), each having two outcomes. The correlations observed by all parties are described by $\{p(a, c, b_1, \dots, b_{N-1} | x, z, y_1, \dots, y_{N-1})\}$, where $z \in \{0, 1, 2\}$ and $c \in \{0, 1\}$ denote the input and output of Charlie, respectively. The first two of Charlie's measurements, labeled by $z = 0, 1$, are performed on Charlie's subsystem of $|\phi_{AC}\rangle$, whereas the third is performed on both of Charlie's subsystems. Let us now go through the steps in detail.

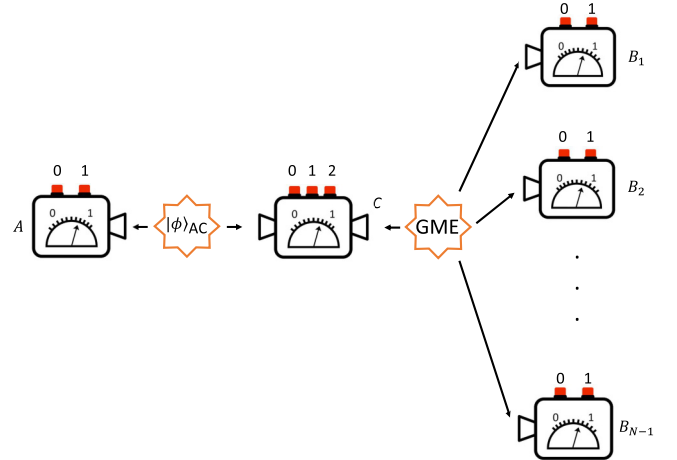


FIG. 3. Extended Bell scenario. An additional party, Charlie, enforces trust in Alice by certifying his measurements. In the considered scenario, there are two sources of states: one between Alice and Charlie distributing bipartite states $|\phi\rangle_{AC}$, and one between Charlie and the $N-1$ Bobs distributing N partite states.

A. Step I: Certifying Alice's observables

To certify Alice's observables, Alice and Charlie simply play the CHSH game [49] on the state distributed by the bipartite source. If the correlations they observe $\{p(a, c|x, y)\}$ violate the CHSH Bell inequality maximally, that is, they achieve the maximal quantum value of the functional

$$I_{\text{CHSH}} := \langle A_0 C_0 \rangle + \langle A_0 C_1 \rangle + \langle A_1 C_0 \rangle - \langle A_1 C_1 \rangle, \quad (34)$$

which is $I_{\text{CHSH}} = 2\sqrt{2}$, then they infer that there exist local unitary operations U_A and U_C such that

$$U_A \otimes U_C |\psi\rangle_{AC} = |\phi^+\rangle_{A'C'} \otimes |\xi\rangle_{A''C''}, \quad (35)$$

where $|\xi\rangle_{A''C''}$ is some auxiliary state, and

$$U_A A_0 U_A^\dagger = Z \otimes \mathbb{1}_{A''}, \quad U_B A_1 U_B^\dagger = X \otimes \mathbb{1}_{B''}. \quad (36)$$

Thus, Alice, by playing the CHSH game with Charlie, can certify her measurements to be, up to the above equivalences, the two Pauli observables that are used in the steering inequality (31).

B. Step II: Teleporting ρ_A from Charlie to Alice

Once the state between Alice and Charlie is certified to be maximally entangled, the local state ρ_A of the multipartite state ρ_{AB} is teleported from Charlie to Alice using Charlie's third measurement corresponding to $z = 2$, which is performed on both systems received by Charlie. We restrict our analysis to only the first outcome of the Bell measurement, as no communication is allowed between Alice and Bob during the protocol. The reason for this teleportation step rather than directly sending the state ρ_A to Alice is that the measurements of Alice are only certified on the local support of $\text{Tr}_C |\phi\rangle\langle\phi|_{AC}$. If the local support is different, then the measurement cannot be guaranteed (see Ref. [31]). Consequently, in the next step, we are only interested in the correlations $\{p(a, 0, b_1, \dots, b_N | x, 2, y_1, \dots, y_N)\}$, keeping in mind the self-tested state and measurements in the previous step.

C. Step III: Self-testing with certified measurements

In the last step, Alice, together with all Bobs, can now use the 1SDI certification methods developed in the previous section to certify a given multipartite state in ρ_{AB} ; however, now, the trusted observables Z and X must be replaced by those given in (36), which have been previously certified by Alice and Charlie.

Consider now an extended Bell-scenario operator $\bar{S}$ constructed, for instance, from the steering operator (28), in which Z and X have been replaced by the measurements

A_1 and A_2 certified in (35). It is not difficult to observe that the new steering operator can be written as

$$\bar{S} = U_A^\dagger (\mathcal{S} \otimes \mathbb{1}_{A''}) U_A, \quad (37)$$

where $\mathcal{S}$ are any of the above-introduced steering operators. Consequently, the LHS and quantum bounds remain the same in the extended-Bell and standard steering scenarios. Moreover, if $\langle \bar{S} \rangle = \beta_Q$, then we readily have $\langle \mathcal{S} \otimes \mathbb{1}_{A''} \rangle = \beta_Q$. Consequently, any state $|\psi_{AB}\rangle$ attaining the maximal value of $\bar{S}$ must also satisfy $\text{Tr}(\mathcal{S} \rho_{A'B}) = \beta_Q$, where $\rho_{A'B} = \text{Tr}_{A''} (U_A \psi_{AB})$. This is the exact condition that we use in the above 1SDI certification proofs, and it can thus be straightforwardly used here to obtain

$$U_A \otimes U_B |\psi_{ABE}\rangle = |\tilde{\psi}_{A'B'}\rangle \otimes |\xi_{B''E}\rangle, \quad (38)$$

where $|\psi'\rangle_{AB}$ is the ideal state and the system E is added to purify $\rho_{A'B}$. Consequently, the state generated by the source is certified to be $U_A \otimes U_B |\psi_{ABE}\rangle = |\tilde{\psi}_{A'B'}\rangle \otimes |\xi_{B''E}\rangle$.

V. DISCUSSION

In this work, we showed that while designing self-testing methods with a minimal number of measurements is highly challenging, introducing trust in one of the parties enables the construction of 1SDI schemes for a broad class of GME states. We focused on graph states, Schmidt states, and generalized W states. Another important finding is that using an arbitrarily low amount of entanglement, it is possible to certify, in a single process, a pair of mutually unbiased operators for an arbitrary number of observers. Inspired by the results of Ref. [20], we also lifted the 1SDI certification to DI self-testing by including an additional party to certify the measurement of the trusted party. While in this work we have only considered the case in which the trusted party performs two-outcome measurements, this technique can be straightforwardly generalized to a higher number of outcomes with the trusted party, provided a suitable Bell inequality exists that allows one to self-test the measurements of the trusted party. One can also use the network-based self-testing of arbitrary measurements presented in Ref. [21] to certify Alice's measurements; however, such an approach would be highly complicated and difficult to implement. We believe that our results open a path toward DI self-testing strategies based on a minimal number of measurements per party that are applicable to any genuinely entangled state. At the moment, no such scheme exists.

Our work raises several follow-up questions. An obvious one is how to expand our approach to certify other classes of genuinely entangled multipartite states. In fact, it would be interesting to see whether the assumption considered here, that one or a few of the measurement devices

are trusted, allows one to design a certification scheme for any GME state based on the minimal number of measurements per observer. A related question that arises here is whether the ideas presented above can be combined with the results of Ref. [50,51] to design self-testing schemes for multipartite states that are based on finite statistics. Furthermore, it would have been interesting to explore whether the ISDI certification scheme for graph states of any local dimension provided here could be generalized to stabilizer subspaces (cf. Ref. [52]). At the same time, as far as the experimental testing of our schemes is concerned, it is crucial to understand how resilient they are to experimental errors. Additionally, our self-testing results can be used for secure multipartite randomness generation.

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DATA AVAILABILITY

No data were created or analyzed in this study.

APPENDIX A: SELF-TESTING ANY GRAPH STATE

Consider a connected graph G with N vertices and edges denoted as e , with their multiplicity denoted by $\gamma_{i,j}$. The graph state $|G\rangle$ corresponding to graph G is then the unique state that stabilizes the following N operators:

$$S(G)_{d,i} = X_i \otimes \bigotimes_{j \in n(i)} Z_j^{\gamma_{i,j}}, \quad i = 1, 2, \dots, N. \quad (\text{A1})$$

Let us now recall the steering operator introduced in the main text:

$$\begin{aligned} \mathcal{I}_d(G, N) = & X_A \otimes \bigotimes_{j \in n(0)} (B_{j,0})^{\gamma_{0,j}} \\ & + \sum_{j \in n(0)} Z_A^{\gamma_{j,0}} \otimes B_{j,1} \otimes \bigotimes_{j' \in n(j) \setminus \{0\}} (B_{j',0})^{\gamma_{jj'}} \\ & + \sum_{j \notin n(0) \cup 0} B_{j,1} \otimes \bigotimes_{j' \in n(j)} (B_{j',0})^{\gamma_{jj'}} + \text{H.c.}, \end{aligned} \quad (\text{A2})$$

where H.c. denotes the Hermitian conjugate of the operator on its left-hand side. Note also that in the above operator $\gamma_{i,j} = \gamma_{j,i}$ for any i and j , where $\gamma_{i,j}$ is the number of edges connecting vertices i and j , and $n(j)$ is the set of vertices connected to vertex j . The cardinality of $n(j)$ is denoted by $\bar{n}_j$. Now, the steering inequality is given by

$$\langle \mathcal{I}_d(G, N) \rangle \leq \beta_L. \quad (\text{A3})$$

The classical bound β_L is given below.

Fact 1. The local-hidden-state bound β_L of the steering functional $\langle \mathcal{I}_d(G, N) \rangle$, where $\mathcal{I}_d(G, N)$ is given in (A2), is upper bounded by

$$\beta_L \leq \max_{\rho} \left(2|\langle X_A \rangle_{\rho}| + 2 \sum_{j \in n(0)} |\langle Z_A^{\gamma_{j,0}} \rangle_{\rho}| \right) + 2(N - \bar{n}_0 - 1), \quad (\text{A4})$$

where Z and X are given in Eq. (1). Furthermore, $\beta_L < 2N$.

Proof. As shown in Ref. [30], for any LHS model with trusted Alice, the expectation values of the joint observables take the form

$$\langle A_x^k \otimes B_y^l \rangle_{\text{LHS}} = \sum_{\lambda} p(\lambda) \langle A_x^k \rangle_{\rho_{\lambda}} \langle B_y^l \rangle_{\lambda}. \quad (\text{A5})$$

Consequently, we can express the steering inequality $\langle \mathcal{I}_d(G, N) \rangle$, where $\mathcal{I}_d(G, N)$ is given in (A2), for any LHS model as

$$\langle \mathcal{I}_d(G, N) \rangle_{\text{LHS}} = \sum_{\lambda} p(\lambda) \langle \mathcal{I}_d(G, N) \rangle_{\lambda, \text{LHS}}, \quad (\text{A6})$$

where $\langle \mathcal{I}_d(G, N) \rangle_{\lambda, \text{LHS}}$ is

$$\begin{aligned} \langle \mathcal{I}_d(G, N) \rangle_{\lambda, \text{LHS}} = & \langle X_A \rangle_{\rho_{\lambda}} \langle \bigotimes_{j \in n(0)} (B_{j,0})^{\gamma_{0,j}} \rangle_{\lambda} + \sum_{j \in n(0)} \langle Z_A^{\gamma_{j,0}} \rangle_{\rho_{\lambda}} \langle B_{j,1} \otimes \bigotimes_{j' \in n(j) \setminus \{0\}} (B_{j',0})^{\gamma_{jj'}} \rangle_{\lambda} \\ & + \sum_{j \notin n(0) \cup 0} \langle B_{j,1} \rangle_{\rho_{\lambda}} \langle \bigotimes_{j' \in n(j)} (B_{j',0})^{\gamma_{jj'}} \rangle_{\lambda} + \langle \text{H.c.} \rangle_{\text{LHS}}. \end{aligned} \quad (\text{A7})$$

As $\langle B_y^l \rangle_\lambda \leq 1$, we obtain

$$\begin{aligned} & \langle \mathcal{I}_d(G, N) \rangle_{\text{LHS}} \\ & \leq \sum_{\lambda} p(\lambda) \left(2|\langle X_A \rangle_{\rho_\lambda}| + 2 \sum_{j \in n(0)} |\langle Z_A^{\gamma_{j,0}} \rangle_{\rho_\lambda}| \right) \\ & \quad + 2(N - \bar{n}_0 - 1), \end{aligned} \quad (\text{A8})$$

which can be upper bounded as

$$\begin{aligned} & \langle \mathcal{I}_d(G, N) \rangle_{\text{LHS}} \\ & \leq \sum_{\lambda} p(\lambda) \max_{\rho} \left(2|\langle X_A \rangle_{\rho}| + 2 \sum_{j \in n(0)} |\langle Z_A^{\gamma_{j,0}} \rangle_{\rho}| \right) \\ & \quad + 2(N - \bar{n}_0 - 1). \end{aligned} \quad (\text{A9})$$

Using the fact that $\sum_{\lambda} p(\lambda) = 1$, the above finally gives us

$$\begin{aligned} \langle \mathcal{I}_d(G, N) \rangle_{\text{LHS}} & \leq \max_{\rho} \left(2|\langle X_A \rangle_{\rho}| + 2 \sum_{j \in n(0)} |\langle Z_A^{\gamma_{j,0}} \rangle_{\rho}| \right) \\ & \quad + 2(N - \bar{n}_0 - 1). \end{aligned} \quad (\text{A10})$$

One can numerically evaluate the above quantity for any d and any graph. Let us finally observe from the above expression that the LHS bound of the steering functional $\mathcal{I}_d(G, N)$ satisfies $\beta_L < 2N$ because Z and X are mutually incompatible and there is no quantum state that can simultaneously stabilize Z and X . ■

Let us now compute the quantum bound of the steering functional $\langle \mathcal{I}_d(G, N) \rangle$.

Fact 2. The quantum bound of the steering functional $\langle \mathcal{I}_d(G, N) \rangle$, where $\mathcal{I}_d(G, N)$ is given in (A2), is given by $\beta_Q = 2N$ and is achieved by the graph state $|G\rangle$ with the observables

$$B_{j,0} = Z, \quad B_{j,1} = X. \quad (\text{A11})$$

Proof. Let us first observe that the number of terms in the steering operator (A2) is equal to twice the number of stabilizers $S(G)_{d,i}$ (13), that is, $2N$. Since the expectation value of each term is upper bounded by 1, we have $\langle \mathcal{I}_d(G, N) \rangle \leq 2N$.

Consider now the unknown observables $B_{i,j}$ given by

$$B_{j,0} = Z, \quad B_{j,1} = X, \quad (\text{A12})$$

along with the graph state $|G\rangle$, which is uniquely stabilized by the $S(G)_i$ operators. It is straightforward to observe that, using these states and observables, one can attain the quantum bound. ■

Now notice that the steering operator consists of $2N$ terms and, consequently, the algebraic bound of $\langle \mathcal{I}_d(G, N) \rangle$ is also $2N$, as all the operators satisfy $B_{i,j}^\dagger B_{i,j} = \mathbb{1}$. Thus, it is necessary that any state $|\psi_N\rangle$ and observables $B_{i,j}$ that achieve the quantum bound must satisfy the following conditions:

$$\langle \psi_N | \bar{S}(G)_i | \psi_N \rangle = 1, \quad i = 1, 2, \dots, N, \quad (\text{A13})$$

where $\bar{S}(G)_i$ represents each term of the steering operator (A2). Now, using the Cauchy-Schwarz inequality, one can conclude from here that

$$\forall i \quad \bar{S}(G)_i | \psi_N \rangle = | \psi_N \rangle. \quad (\text{A14})$$

The above condition would be extremely useful for certifying any graph state. Let us now establish the following theorem.

Theorem A1. Assume that the steering inequality (A3) is maximally violated when the 0th party, Alice, chooses the observables A_i defined to be $A_0 = Z$ and $A_1 = X$, and all the other parties choose their observables as $B_{i,j}$ for $i = 1, 2, \dots, N-1$ and $j \in \{0, 1\}$ acting on $\mathcal{H}_{B_i}$. Let us say that the state that attains this violation is given by $|\psi_N\rangle_{AB} \in \mathbb{C}^d \otimes \bigotimes_{i=1}^{N-1} \mathcal{H}_{B_i} \otimes \mathcal{H}_E$. Then, the following statement holds true for any d : $\mathcal{H}_{B_i} = \mathcal{H}_{B'_i} \otimes \mathcal{H}_{B''_i}$, where $\mathcal{H}_{B'_i} \equiv \mathbb{C}^d$, $\mathcal{H}_{B''_i}$ is some finite-dimensional Hilbert space, and there exists a local unitary transformation on Bob's side $U_i : \mathcal{H}_{B_i} \rightarrow \mathcal{H}_{B_i}$, such that

$$\forall j, \quad U_i B_{j,0} U_i^\dagger = Z_{B'_i} \otimes \mathbb{1}_{B''_i}, \quad U_i B_{j,1} U_i^\dagger = X_{B'_i} \otimes \mathbb{1}_{B''_i}, \quad (\text{A15})$$

where B'' denotes Bob's auxiliary system and the state $|\psi_N\rangle$ is given by

$$\left(\mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_i \right) |\psi_N\rangle_{ABE} = |G\rangle_{AB'} \otimes |\xi\rangle_{B''E}, \quad (\text{A16})$$

where $|\xi\rangle_{B''E} \in \bigotimes_{i=1}^{N-1} \mathcal{H}_{B''_i} \otimes \mathcal{H}_E$ denotes the auxiliary state.

Proof. Let us begin by considering the first and second terms of the steering operator (A2) corresponding to the stabilizers when choosing the 0th vertex and vertices connected to it, respectively. As concluded in Eq. (A14),

we have

$$X_A \otimes \bigotimes_{j \in n(0)} (B_{j,0})^{\gamma_{0,j}} |\psi_N\rangle = |\psi_N\rangle \quad (\text{A17})$$

and

$$Z_A^{\gamma_{j,0}} \otimes B_{j,1} \otimes \bigotimes_{j' \in n(j) \setminus \{0\}} (B_{j',0})^{\gamma_{j,j'}} |\psi_N\rangle = |\psi_N\rangle \quad \forall j \in n(0). \quad (\text{A18})$$

Let us now choose a particular $k \in n(0)$ and rewrite the above conditions as

$$X_A \otimes (B_{k,0})^{\gamma_{0,k}} \otimes \bigotimes_{j \in n(0) \setminus \{k\}} (B_{j,0})^{\gamma_{0,j}} |\psi_N\rangle = |\psi_N\rangle \quad (\text{A19})$$

and

$$Z_A^{\gamma_{k,0}} \otimes B_{k,1} \otimes \bigotimes_{j' \in n(k) \setminus \{0\}} (B_{j',0})^{\gamma_{k,j'}} |\psi_N\rangle = |\psi_N\rangle. \quad (\text{A20})$$

Since the observables X and Z are unitary and $\gamma_{0,k} = \gamma_{k,0}$, we have

$$(B_{k,0})^{\gamma_{0,k}} \otimes \bigotimes_{j \in n(0) \setminus \{k\}} (B_{j,0})^{\gamma_{0,j}} |\psi_N\rangle = X_A^{-1} |\psi_N\rangle \quad (\text{A21})$$

and

$$B_{k,1} \otimes \bigotimes_{j' \in n(k) \setminus \{0\}} (B_{j',0})^{\gamma_{k,j'}} |\psi_N\rangle = Z_A^{-\gamma_{0,k}} |\psi_N\rangle. \quad (\text{A22})$$

Now, multiplying $Z_A^{-\gamma_{0,k}}$ to the condition (A21) and then using Eq. (A22), we get

$$(B_{k,0})^{\gamma_{0,k}} B_{k,1} \otimes \left(\bigotimes_{j \in n(0) \setminus \{k\}} (B_{j,0})^{\gamma_{0,j}} \right) \left(\bigotimes_{j' \in n(k) \setminus \{0\}} (B_{j',0})^{\gamma_{k,j'}} \right) \times |\psi_N\rangle = Z_A^{-\gamma_{0,k}} X_A^{-1} |\psi_N\rangle. \quad (\text{A23})$$

Then, multiplying condition (A22) by X_A^{-1} and using Eq. (A21), we get

$$B_{k,1} (B_{k,0})^{\gamma_{0,k}} \otimes \left(\bigotimes_{j' \in n(k) \setminus \{0\}} (B_{j',0})^{\gamma_{k,j'}} \right) \left(\bigotimes_{j \in n(0) \setminus \{k\}} (B_{j,0})^{\gamma_{0,j}} \right) \times |\psi_N\rangle = X_A^{-1} Z_A^{-\gamma_{0,k}} |\psi_N\rangle. \quad (\text{A24})$$

Note that terms inside the large brackets in both the above formulas commute. Now, using the fact that

$\omega^{\gamma_{0,k}} Z_A^{-\gamma_{0,k}} X_A^{-1} = X_A^{-1} Z_A^{-\gamma_{0,k}}$, we can multiply (A23) by $\omega^{\gamma_{0,k}}$ and subtract (A24) to obtain

$$\begin{aligned} & (\omega^{\gamma_{0,k}} (B_{k,0})^{\gamma_{0,k}} B_{k,1} - B_{k,1} (B_{k,0})^{\gamma_{0,k}}) \\ & \otimes \left(\bigotimes_{j \in n(0) \setminus \{k\}} (B_{j,0})^{\gamma_{0,j}} \right) \left(\bigotimes_{j' \in n(k) \setminus \{0\}} (B_{j',0})^{\gamma_{k,j'}} \right) \\ & \times |\psi_N\rangle = 0. \end{aligned} \quad (\text{A25})$$

As the terms inside the large brackets in the above expression are unitary and thus invertible, we arrive at a simple condition:

$$(\omega^{\gamma_{0,k}} (B_{k,0})^{\gamma_{0,k}} B_{k,1} - B_{k,1} (B_{k,0})^{\gamma_{0,k}}) \otimes \mathbb{1}_{A\mathbf{B}E \setminus \{B_k\}} |\psi_N\rangle = 0. \quad (\text{A26})$$

By taking a partial trace over all the subsystems except the k th one, we get

$$(\omega^{\gamma_{0,k}} (B_{k,0})^{\gamma_{0,k}} B_{k,1} - B_{k,1} (B_{k,0})^{\gamma_{0,k}}) \rho_{k,N} = 0, \quad (\text{A27})$$

where $\rho_{k,N} = \text{Tr}_{A\mathbf{B}E \setminus \{B_k\}}(|\psi_N\rangle\langle\psi_N|)$. As discussed before, the local states of each party are full rank and thus invertible. Consequently, we finally have

$$\omega^{\gamma_{0,k}} (B_{k,0})^{\gamma_{0,k}} B_{k,1} = B_{k,1} (B_{k,0})^{\gamma_{0,k}}. \quad (\text{A28})$$

As proven in Ref. [13], if two unitary observables $B_{k,0}$ and $B_{k,1}$, for which $B_{k,0}^d = B_{k,1}^d = \mathbb{1}$, satisfy relation (A28), then there exists a unitary $U_k : \mathcal{H}_{B_k} \rightarrow \mathcal{H}_{B'_k}$ such that

$$U_k B_{k,0} U_k^\dagger = Z_{B'_k} \otimes \mathbb{1}_{B''_k}, \quad U_k B_{k,1} U_k^\dagger = X_{B'_k} \otimes \mathbb{1}_{B''_k}. \quad (\text{A29})$$

Similarly, we can reach the above conclusion for any vertex $j \in n(0)$. Using these derived observables, we then find the observables for vertices connected with any vertex $j \in n(0)$ in exactly the same manner as above. Since any vertex is connected to at least one other vertex, we continue this until all the observables are found and we get

$$U_i B_{i,0} U_i^\dagger = Z_{B'_i} \otimes \mathbb{1}_{B''_i}, \quad U_i B_{i,1} U_i^\dagger = X_{B'_i} \otimes \mathbb{1}_{B''_i} \quad \forall i. \quad (\text{A30})$$

Let us now characterize the state $|\psi_N\rangle$ that achieves the quantum bound of the steering functional $\langle \mathcal{I}_d(G, N) \rangle$ (A2). Notice that by inserting the derived observables from (A30) into the condition (A14), we get

$$\forall i \quad \bar{S}(G)_i |\psi_N\rangle = \left(\bigotimes_{i=1}^{N-1} U_i^\dagger \right) S(G)_{d,i,AB'_1 \dots B'_N} \otimes \mathbb{1}_{B'_1 \dots B'_N E} \\ \times \left(\bigotimes_{i=1}^{N-1} U_i \right) |\psi_N\rangle = |\psi_N\rangle, \quad (\text{A31})$$

where $S(G)_{d,i}$ are the stabilizing operators of a graph state $|G\rangle$ of the form (13). As discussed above, the graph state $|G\rangle$ is the unique eigenstate of such operators with eigenvalue 1. Consequently, we have

$$\left(\bigotimes_{i=1}^{N-1} U_i \right) |\psi_N\rangle_{ABE} = |G\rangle_{AB'} \otimes |\xi\rangle_{B'E}, \quad (\text{A32})$$

where $|\xi\rangle_{B'E} \in \bigotimes_{i=1}^{N-1} \mathcal{H}_{B'_i} \otimes \mathcal{H}_E$ denotes the auxiliary state. ■

Let us now restrict ourselves to $d = 2$ and find the robust self-testing statement. Before proceeding, let us state a general Lemma that will be required for the robust proof.

Lemma A1. Consider two Hermitian and unitary matrices A_0 and A_1 acting on one of the subsystems of a bipartite state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$. Assume then that $\| \{A_0, A_1\} |\psi\rangle \| \leq \varepsilon_N$ for some $0 < \varepsilon < 1$, where we additionally assume that, under Jordan's lemma, both A_0 and A_1 consist of only nontrivial 2×2 blocks. Then, there exists a local unitary transformation $U : \mathcal{H}_A \rightarrow \mathbb{C}^2 \otimes \mathcal{H}_{A'}$ such that

$$UAU^\dagger |\psi'\rangle = Z \otimes \mathbb{1} |\psi'\rangle, \\ \|(UBU^\dagger - X \otimes \mathbb{1})|\psi'\rangle\| \leq 2\varepsilon, \quad (\text{A33})$$

where $|\psi'\rangle = U|\psi\rangle$, and for simplicity, we have omitted all the identities acting on the second subsystem B .

The proof of this lemma can be found in Appendix B of Ref. [53]. Let us now derive the robustness statement.

Theorem A2. Let us consider a graph state $|G\rangle$ that we want to certify robustly. For this graph, we consider a hierarchy of nodes as follows: (0) $n(0)$: all nodes connected to the 0th node (trusted node), (1) $n_1(j)$: all nodes connected to some j th node such that $j \in n(0)$, which are not neighbors of the node 0, nor the node 0 itself, (2) $n_2(j)$: all nodes connected to some node in $n_1(j)$ such that they are not connected to 0 and $n(j)$, etc. Since we consider graphs that are not disjoint, for each vertex there exists a path connecting it to the trusted node.

Assume now that the steering inequality (A3) attains a value close to the maximal violation, $\langle \mathcal{I}_2(G, N) \rangle \geq \beta_Q - \varepsilon$, when the 0th party (trusted) Alice chooses the observables A_i defined to be $A_0 = Z$ and $A_1 = X$. Then, the state $|\psi\rangle_{AB}$ and the observables $B_{i,j}$ for $i = 1, \dots, N-1$ and $j = 0, 1$ are certified as

$$\| |\psi'\rangle_{ABE} - |G\rangle_{AB'} \otimes |\xi\rangle_{B'E} \| \leq \frac{2\sqrt{N} + g}{2} \sqrt{\varepsilon}, \quad (\text{A34})$$

where $g = \sum_k n_k a_k$ is the number of nodes connected in the k th hierarchy, $a_k = (8/3)(4^{k+1} - 1)$, and for a node i belonging to the k th hierarchy,

$$U_{B_i} B_{i,0} U_{B_i}^\dagger |\psi'\rangle = Z \otimes \mathbb{1} |\psi'\rangle, \\ \|(U_{B_i} B_{i,1} U_{B_i}^\dagger - X \otimes \mathbb{1})|\psi'\rangle\| \leq a_k \sqrt{\varepsilon}, \quad (\text{A35})$$

where $|\psi'\rangle = \bigotimes_{i=1}^{N-1} U_{B_i} |\psi\rangle$ and we have dropped the indices ABE .

Proof. Let us consider the steering operator $\mathcal{I}_2(G, N)$ given in Eq. (A2), which, in the particular case of $d = 2$, simplifies to

$$\mathcal{I}_2(G, N) = 2 \left[X_A \otimes \bigotimes_{j \in n(0)} B_{j,0} + \sum_{j \in n(0)} Z_A \otimes B_{j,1} \otimes \bigotimes_{j' \in n(j) \setminus \{0\}} B_{j',0} + \sum_{j \notin n(0) \cup 0} B_{j,1} \otimes \bigotimes_{j' \in n(j)} B_{j',0} \right], \quad (\text{A36})$$

where the factor 2 is a consequence of the fact that all local observables are Hermitian in this case.

Let us then observe that it admits the following sum-of-squares decomposition, $\beta_Q \mathbb{1} - \mathcal{I}_2(G, N) = \sum_{j=1}^N P_j P_j^\dagger$, where the operators P_j are given by

$$P_0 = \mathbb{1} - X_A \otimes \bigotimes_{i \in n(0)} B_{i,0}, \\ P_j = \mathbb{1} - Z_A \otimes B_{j,1} \otimes \bigotimes_{j' \in n(j) \setminus \{0\}} B_{j',0} \quad \text{for } j \in n(0), \\ P_j = \mathbb{1} - B_{j,1} \otimes \bigotimes_{j' \in n(j)} B_{j',0} \quad \text{for } j \notin n(0) \cup 0. \quad (\text{A37})$$

Consequently, when $\langle \mathcal{I}_2(G, N) \rangle \geq \beta_Q - \varepsilon$, we have $\|P_j|\psi\rangle\| \leq \sqrt{\varepsilon}$ for all j .

Let us now choose the k th vertex such that $k \in n(0)$, take two of the above relations, and expand using (A37) as

$$\begin{aligned} \left\| |\psi\rangle - X_A \otimes B_{k,0} \otimes \bigotimes_{j \in n(0) \setminus \{k\}} B_{j,0} |\psi\rangle \right\| &\leq \sqrt{\varepsilon}, \\ \left\| |\psi\rangle - Z_A \otimes B_{k,1} \otimes \bigotimes_{j' \in n(k) \setminus \{0\}} B_{j',0} |\psi\rangle \right\| &\leq \sqrt{\varepsilon}. \end{aligned} \quad (\text{A38})$$

Now, using the fact that X and Z are Hermitian and unitary for $d = 2$, and that the vector norm is unitarily invariant, the above inequalities imply that

$$\begin{aligned} \left\| X_A |\psi\rangle - B_{k,0} \otimes \bigotimes_{j \in n(0) \setminus \{k\}} B_{j,0} |\psi\rangle \right\| &\leq \sqrt{\varepsilon}, \\ \left\| Z_A |\psi\rangle - B_{k,1} \otimes \bigotimes_{j' \in n(k) \setminus \{0\}} B_{j',0} |\psi\rangle \right\| &\leq \sqrt{\varepsilon}. \end{aligned} \quad (\text{A39})$$

Exploiting the unitary invariance of the vector norm once more, we can multiply the expressions under the norm by Z_A and X_A , respectively, to obtain

$$\begin{aligned} \left\| Z_A X_A |\psi\rangle - Z_A \otimes B_{k,0} \otimes \bigotimes_{j \in n(0) \setminus \{k\}} B_{j,0} |\psi\rangle \right\| &\leq \sqrt{\varepsilon}, \\ \left\| X_A Z_A |\psi\rangle - X_A \otimes B_{k,1} \otimes \bigotimes_{j' \in n(k) \setminus \{0\}} B_{j',0} |\psi\rangle \right\| &\leq \sqrt{\varepsilon}. \end{aligned} \quad (\text{A40})$$

Let us now consider the second inequality in (A39), and, using the unitary invariance of the norm as well as the fact that all observables are unitary, let us expand it to the following form:

$$\left\| Z_A \otimes B_{k,0} \otimes \bigotimes_{j \in n(0) \setminus \{k\}} B_{j,0} |\psi\rangle - B_{k,0} B_{k,1} \otimes \left(\bigotimes_{j' \in n(k) \setminus \{0\}} B_{j',0} \right) \left(\bigotimes_{j' \in n(k) \setminus \{0\}} B_{j',0} \right) |\psi\rangle \right\| \leq \sqrt{\varepsilon}. \quad (\text{A41})$$

Then, with the aid of the triangle inequality for the vector norm, one shows that the first inequality in (A40) and (A41) imply that

$$\left\| Z_A X_A |\psi\rangle - B_{k,0} B_{k,1} \otimes \left(\bigotimes_{j \in n(0) \setminus \{k\}} B_{j,0} \right) \left(\bigotimes_{j' \in n(k) \setminus \{0\}} B_{j',0} \right) |\psi\rangle \right\| \leq 2\sqrt{\varepsilon}. \quad (\text{A42})$$

Analogously, by suitably modifying the first inequality in (A39) and combining it with the second inequality in (A40) through the triangle inequality, one can obtain

$$\left\| X_A Z_A |\psi\rangle - B_{k,1} B_{k,0} \otimes \left(\bigotimes_{j \in n(0) \setminus \{k\}} B_{j,0} \right) \left(\bigotimes_{j' \in n(k) \setminus \{0\}} B_{j',0} \right) |\psi\rangle \right\| \leq 2\sqrt{\varepsilon}. \quad (\text{A43})$$

In deriving these inequalities we also exploited the fact that $B_{j,0}$ commute among each other if they act on different sites. Now, using the fact that $Z_A X_A = -X_A Z_A$, and then exploiting the triangle inequality, we have

$$\| [B_{k,0} B_{k,1} + B_{k,1} B_{k,0}] |\psi\rangle \| \leq 4\sqrt{\varepsilon}, \quad k = 1, \dots, n(0). \quad (\text{A44})$$

To all the observables $B_{k,i}$ for $k \in n(0)$ and $i = 0, 1$ (which are those that are measured by the neighbors of the trusted party) we can apply Lemma A1 (where for simplicity we additionally assume that $B_{k,i}$ decompose under Jordan's lemma only into nontrivial 2×2 blocks), to conclude that there exists a local unitary operation U_i acting on the i th node such that

$$\begin{aligned} U_i B_{i,0} U_i^\dagger |\psi'\rangle &= Z \otimes \mathbb{1} |\psi'\rangle, \\ \left\| (U_i B_{i,1} U_i^\dagger - X \otimes \mathbb{1}) |\psi'\rangle \right\| &\leq 8\sqrt{\varepsilon}, \end{aligned} \quad (\text{A45})$$

where $|\psi'\rangle = \bigotimes_{i \in n(0)} U_i |\psi\rangle$.

The above steps allow us to find the robust bounds to observables for the nodes j directly connected to the trusted party $j \in n(0)$. Based on the relations in Eq. (A45), we can follow the same reasoning as above to prove analogous relations for the observables measured by the neighbors of all nodes in $n(0)$ that do not themselves belong to $n(0)$ and are not the trusted party. To this aim, let us now consider one of these nodes $j \in n(0)$ and prove that relations similar to (A45) hold true for all nodes $k \notin n(0) \cup \{0\}$ connected to j . We can follow the same steps as above using the sum-of-squares operators P_j and P_k (A37) to get

$$\begin{aligned} dl|\psi\rangle - B_{k,1} \otimes B_{j,0} \otimes \bigotimes_{j' \in n(k)} B_{j',0} |\psi\rangle &\leq \sqrt{\varepsilon}, \\ \left\| |\psi\rangle - Z_A \otimes B_{j,1} \otimes B_{k,0} \otimes \bigotimes_{j' \in n(j) \setminus \{0,k\}} B_{j',0} |\psi\rangle \right\| &\leq \sqrt{\varepsilon}. \end{aligned} \quad (\text{A46})$$

As the observables $B_{j,0}$ and $B_{j,1}$ for $j \in n(0)$ are certified as in Eq. (A45), this allows us to obtain

$$\begin{aligned} \left\| |\psi'\rangle - B_{k,1} \otimes (Z \otimes \mathbb{1})_j \otimes \bigotimes_{j' \in n(k)} B_{j',0} |\psi'\rangle \right\| &\leq \sqrt{\varepsilon}, \\ \left\| |\psi'\rangle - Z_A \otimes (X \otimes \mathbb{1})_j \otimes B_{k,0} \otimes \bigotimes_{j' \in n(j) \setminus \{0,k\}} B_{j',0} |\psi'\rangle \right\| &\leq 9\sqrt{\varepsilon}. \end{aligned} \quad (\text{A47})$$

Following the same steps from Eqs. (A40) to (A44), we obtain

$$\begin{aligned} \left\| [B_{k,0} B_{k,1} + B_{k,1} B_{k,0}] |\psi\rangle \right\| &\leq 20\sqrt{\varepsilon} \\ (k \in n(j), k \notin n(0) \cup \{0\}). \end{aligned} \quad (\text{A48})$$

This allows us to conclude from Lemma A1 that for each $i \in n(j), i \notin n(0) \cup \{0\}$, there exists a unitary operation U_i

such that

$$\begin{aligned} U_i B_{i,0} U_i^\dagger |\psi''\rangle &= Z \otimes \mathbb{1} |\psi''\rangle, \\ \left\| (U_i B_{i,1} U_i^\dagger - X \otimes \mathbb{1}) |\psi''\rangle \right\| &\leq 40\sqrt{\varepsilon}, \end{aligned} \quad (\text{A49})$$

where $|\psi''\rangle = \bigotimes_{i \in n(j), i \notin n(0) \cup \{0\}} U_i |\psi\rangle$.

We now define a hierarchy of nodes as follows: (0) $n(0)$: all nodes connected to the 0th node (trusted node), (1) $n_1(j)$: all nodes connected to some j th node such that $j \in n(0)$, which are neither neighbors of node 0 nor node 0 itself, (2) $n_2(j)$: all nodes connected to some node in $n_1(j)$ such that they are not connected to 0 and $n(j)$, etc. We proceed in this way until all vertices are accounted for; clearly, since we consider graphs that are not disjoint, for each vertex there exists a path connecting it to the trusted node. At each level, the error in the relations (A45) and (A48) for the observables $B_{i,1}$ accumulates as 8, 40, 168, $\dots$; one can check that this series is described by $a_k = (8/3)(4^{k+1} - 1)$ for the k th level of the hierarchy. More precisely, Bob's observables satisfy

$$\left\| [B_{i,0} B_{i,1} + B_{i,1} B_{i,0}] |\psi\rangle \right\| \leq a_k \sqrt{\varepsilon} \quad (k\text{th level}). \quad (\text{A50})$$

Let us again consider the steering operator $\mathcal{I}_2(G, N)$ (A2) and observe that $2\beta_Q[\beta_Q \mathbb{1} - \mathcal{I}_2(G, N)] = (\beta_Q \mathbb{1} - \mathcal{I}_2(G, N))^2$, where we have used the fact that $\mathcal{I}_2(G, N)^2 = \beta_Q \mathbb{1}$, in which $\beta_Q = 2N$. Consequently, from the assumption that $\langle \mathcal{I}_2(G, N) \rangle \geq \beta_Q - \varepsilon$, we infer the following inequality:

$$\left\| [\beta_Q \mathbb{1} - \mathcal{I}_2(G, N)] |\psi\rangle \right\| \leq 2\sqrt{N\varepsilon}. \quad (\text{A51})$$

As $\mathcal{I}_2(G, N)$ is composed of the observables $B_{i,0}$ and $B_{i,1}$, which are certified as (A45), we can rewrite the above formula using the triangle inequality as

$$\begin{aligned} \left\| [\beta_Q \mathbb{1} - \mathcal{I}_2^{\text{id}} \otimes \mathbb{1}_{B'_1 \dots B'_{N-1}}(G, N)] |\psi'\rangle \right\| \\ \leq 2\sqrt{N\varepsilon} + g\sqrt{\varepsilon} = \mu\sqrt{\varepsilon}, \end{aligned} \quad (\text{A52})$$

where $\mathcal{I}_2^{\text{id}} = 2 \sum_{i=1}^N \bar{S}(G)_{2,i}$ [c.f. Eq. (13)] and $\mu = 2\sqrt{N} + g$, with the factor g accounting for the graph structure so that the errors accumulate according to the hierarchy of nodes defined below Eq. (A49). Suppose that at the k th hierarchy, there are n_k nodes, then $g = \sum_k n_k a_k$.

Consider now the eigensystem of the ideal steering operator, $\mathcal{I}_2^{\text{id}} |G_l\rangle = \lambda_l |G_l\rangle$. It is simple to observe that the difference between the highest and second-highest eigenvalues is 4, as the second-highest eigenvalue is attained by $|G_l\rangle$ where $l_k = 1$ for some k and $l_i = 0$ for all $i \neq k$. Let us now express the above formula (A52) using the

eigensystem of $\mathcal{I}_I^{\text{id}}$ as

$$\left\| \left[\sum_{s=1}^{2^N} (\beta_Q - \lambda_l) |G_l\rangle\langle G_l| \otimes \mathbb{1}_{A_1'' \dots A_N''} \right] |\psi'\rangle \right\| \leq \mu \sqrt{\varepsilon}. \quad (\text{A53})$$

Let us now implement the following expansion of the state $|\psi'\rangle$:

$$|\psi'\rangle = \sum_{i=1}^{2^N} \alpha_i |G_i\rangle |\xi_i\rangle, \quad (\text{A54})$$

where $|G_i\rangle$ are the eigenstates of the ideal steering operator $\mathcal{I}^{\text{id}}$, i.e., $\mathcal{I}^{\text{id}}|G_i\rangle = \lambda_i|G_i\rangle$. Then, α_i are non-negative coefficients such that $\sum_i \alpha_i^2 = 1$, and $|\xi_i\rangle$ are some auxiliary vectors with $|\xi_i\rangle \in \bigotimes_{i=1}^N \mathcal{H}_{B_i''}$. The fact that α_i can be taken as non-negative stems from the freedom in choosing the auxiliary vectors $|\xi_i\rangle$. Let us also enumerate the eigenvalues λ_i so that $\lambda_1 = 2N$, which is the quantum bound β_Q ; then, the corresponding eigenstate $|G_1\rangle$ is the graph state to be self-tested, i.e., $|G_1\rangle \equiv |G\rangle$.

Using this decomposition, the expression (A53) allows one to deduce that

$$\left\| \sum_{l=1}^{2^N} (\beta_Q - \lambda_l) \alpha_l |G_l\rangle |\xi_l\rangle \right\| \leq \mu \sqrt{\varepsilon}, \quad (\text{A55})$$

which, on expanding the left-hand side, results in

$$\sum_{l=1}^{2^N} \alpha_l^2 (\beta_Q - \lambda_l)^2 \leq \mu^2 \varepsilon. \quad (\text{A56})$$

Let us now consider the eigenvalues λ_i of the ideal steering operator $\mathcal{I}^{\text{id}}$. Clearly, the largest eigenvalue is exactly the quantum bound $\beta_Q = 2N$, and the corresponding term in the above sum therefore vanishes. It is then not difficult to observe that the next-largest eigenvalue is $2N - 4$, which allows us to write the following inequality:

$$\sum_{l=1}^{2^N} \alpha_l^2 (\beta_Q - \lambda_l)^2 \geq 4 \sum_{l=2}^{2^N} \alpha_l^2, \quad (\text{A57})$$

where we used the fact that $(\beta_Q - \lambda_l)^2 \geq 4$ for every $l = 2, \dots, 2^N$. Consequently, we have the following inequality:

$$\sum_{l=2}^{2^N} \alpha_l^2 \leq \mu^2 \frac{\varepsilon}{4}. \quad (\text{A58})$$

We then use the fact that $\sum_i \alpha_i^2 = 1$, which, together with (A58), implies that $\alpha_1 \geq \alpha_1^2 \geq 1 - (18N)^2 \frac{\varepsilon}{4}$. Thus,

$$\langle \psi' | (|G\rangle \langle \xi_1|) | \psi' \rangle = \alpha_1 \geq 1 - \mu^2 \frac{\varepsilon}{4}, \quad (\text{A59})$$

which directly implies that

$$\begin{aligned} \left\| |\psi'\rangle - |G\rangle \langle \xi_1| \right\| &= \sqrt{2[1 - \text{Re}[\langle \psi' | (|G\rangle \langle \xi_1|) | \psi' \rangle]]} \\ &= \sqrt{2(1 - \alpha_1)} \leq \frac{\mu}{2} \sqrt{\varepsilon}, \end{aligned} \quad (\text{A60})$$

which is the desired inequality (A34). This completes the proof. $\blacksquare$

APPENDIX B: SELF-TESTING SCHMIDT STATES

The Schmidt states we consider have the form

$$|\psi(\boldsymbol{\alpha})\rangle_{AB} = \sum_{i=0}^{d-1} \alpha_i |i\rangle^{\otimes N}, \quad (\text{B1})$$

where $d = \min\{d_A, d_{B_i}\}$ and $\boldsymbol{\alpha}$ is a vector composed of the Schmidt coefficients α_i , such that $0 < \alpha_i < 1$ for all i and $\sum_{i=0}^{d-1} \alpha_i^2 = 1$. Next, we are going to introduce a steering operator that provides an inequality maximally violated by states of the form (B1).

Let us now recall the steering operator for Schmidt states introduced in the main text:

$$\begin{aligned} \mathcal{S}_d(\boldsymbol{\alpha}, N) &= \sum_{k=1}^{d-1} \left[\left(\sum_{j=1}^{N-1} A_0^k \otimes B_{j,0}^k \right) + \gamma(\boldsymbol{\alpha}) A_1^k \right. \\ &\quad \left. \otimes \bigotimes_{j=1}^{N-1} B_{j,1}^k + \delta_k(\boldsymbol{\alpha}) A_0^k \right], \end{aligned} \quad (\text{B2})$$

where

$$\gamma(\boldsymbol{\alpha}) = \frac{d}{\sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j}}, \quad \delta_k(\boldsymbol{\alpha}) = -\frac{\gamma(\boldsymbol{\alpha})}{d} \sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j} \omega^{k(d-j)}. \quad (\text{B3})$$

In our scenario, we consider that all the $N - 1$ Bobs perform two arbitrary measurements denoted by B_{i_0} and B_{i_1} . We also assume that the local dimensions of the state they share are unknown. At the same time, because Alice's device is trusted, her measurements are completely characterized, and the local dimension of the shared state on Alice's side is known. Alice's observables are the generalized Pauli Z and X operators,

$$Z_d = \sum_{i=0}^{d-1} \omega^i |i\rangle \langle i|, \quad X_d = \sum_{i=0}^{d-1} |i+1\rangle \langle i|. \quad (\text{B4})$$

The corresponding steering inequality is given by

$$\langle \mathcal{S}_d(\boldsymbol{\alpha}, N) \rangle \leq \beta_L, \quad (\text{B5})$$

where β_L is the LHS bound given below.

Fact 3. The local-hidden-state bound β_L of the steering functional $\langle \mathcal{S}_d(\boldsymbol{\alpha}) \rangle$, where $\mathcal{S}_d(\boldsymbol{\alpha})$ is given in (B2), is upper bounded by

$$\beta_L \leq \max_{\substack{|\eta_0|, \dots, |\eta_{d-1}| \\ |\eta_0|^2 + \dots + |\eta_{d-1}|^2 = 1}} \left(d(N-1) \max_a |\eta_a|^2 + \gamma(\boldsymbol{\alpha}) \left[\left(\sum_{i=0}^{d-1} |\eta_i| \right)^2 - \sum_{i=0}^{d-1} \alpha_i \sum_{a=0}^{d-1} \frac{|\eta_a|^2}{\alpha_a} \right] \right) - (N-2). \quad (\text{B6})$$

Furthermore, $\beta_L < (d-1)(N-1) + 1$.

Proof. To determine the LHS bound β_L of the inequality in Eq. (B2), we first express the functional $\mathcal{S}_d(\boldsymbol{\alpha}, N)$ in the probability picture as follows:

$$\begin{aligned} \mathcal{S}_d(\boldsymbol{\alpha}) &= d \sum_{j=1}^{N-1} \sum_{a, b_j=0}^{d-1} c_{a, b_j} p(a, b_j | 0, 0) + \gamma(\boldsymbol{\alpha}) \left(d \sum_{a, \mathbf{b}=0}^{d-1} c_{a, \mathbf{b}} p(a, \mathbf{b} | 1, 1, \dots, 1) - \sum_{\substack{i, a=0 \\ i \neq a}}^{d-1} \alpha_i \frac{p(a|0)}{\alpha_a} \right) \\ &\quad - (N-1) - \gamma(\boldsymbol{\alpha}) - \delta_0(\boldsymbol{\alpha}), \end{aligned} \quad (\text{B7})$$

where $c_{a, b} = 1$ if $a \oplus_d b = 0$ and 0 otherwise, where $\oplus_d$ is the addition modulo d . Further, $c_{a, \mathbf{b}} = 1$ if $a \oplus_d b_1 \oplus_d \dots \oplus_d b_{N-1} = 0$ and 0 otherwise. We implement this notation because $\sum_{k=0}^{d-1} \omega^{kn(a-b)} = d\delta_{a,b}$ for any n . Likewise, $\sum_{k=0}^{d-1} \omega^{k(d+a-b)} = d\delta_{a,b}$. We note from Eq. (B3) that $\delta_0(\boldsymbol{\alpha}) = -1$, which results in

$$\mathcal{S}_d(\boldsymbol{\alpha}) = d \sum_{j=1}^{N-1} \sum_{a, b_j=0}^{d-1} c_{a, b_j} p(a, b_j | 0, 0) + \gamma(\boldsymbol{\alpha}) \left(d \sum_{a, \mathbf{b}=0}^{d-1} c_{a, \mathbf{b}} p(a, \mathbf{b} | 1, 1, \dots, 1) - \sum_{i=0}^{d-1} \alpha_i \sum_{a=0}^{d-1} \frac{p(a|0)}{\alpha_a} \right) - (N-2). \quad (\text{B8})$$

We consider the LHS model in which the correlations are given as

$$p(a, \mathbf{b} | x, \mathbf{y}) = \sum_{\lambda \in \Lambda} p(\lambda) p(a | x, \rho_A^\lambda) p(\mathbf{b} | \mathbf{y}, \lambda), \quad (\text{B9})$$

where Λ is some set of hidden variables λ shared between Alice and all the Bobs with probability distribution $p(\lambda)$, whereas $p(a | x, \rho_A^\lambda)$ and $p(\mathbf{b} | \mathbf{y}, \lambda)$ are local probability distributions. Recall that Alice's distributions are quantum and depend on the hidden state ρ_A^λ . The LHS model transforms our steering functional as

$$\begin{aligned} \mathcal{S}_d(\boldsymbol{\alpha}) &= d \sum_{j=1}^{N-1} \sum_{a=0}^{d-1} \sum_{\lambda \in \Lambda} p(\lambda) p(a | 0, \rho_A^\lambda) p(d-a | 0, \lambda) + \gamma(\boldsymbol{\alpha}) \left(d \sum_{a=0}^{d-1} \sum_{\lambda \in \Lambda} p(\lambda) p(a | 1, \rho_A^\lambda) p(\mathbf{b} | 1, \dots, 1, \lambda) \right. \\ &\quad \left. - \sum_{i=0}^{d-1} \alpha_i \sum_{a=0}^{d-1} \frac{p(a|0, \rho_A)}{\alpha_a} \right) - (N-2), \end{aligned} \quad (\text{B10})$$

where $b_j = d-a$ for every j in the first term, $a \oplus_d b_1 \oplus_d \dots \oplus_d b_{N-1} = 0$ in the second term, and in the last term, we have used the fact that Alice's distributions are quantum and substituted $\rho_A = \sum_{\lambda} \rho_A^\lambda$. Now, we note that the first two terms in the above expression are bounded from above in the following way:

$$\begin{aligned}
\sum_{a=0}^{d-1} \sum_{\lambda \in \Lambda} p(\lambda) p(a|0, \rho_A^\lambda) p(d-a|0, \lambda) &\leq \sum_{\lambda \in \Lambda} p(\lambda) \max_a p(a|0, \rho_A^\lambda) \leq \max_{|\psi\rangle \in \mathcal{C}^d} \max_a p(a|0, |\psi\rangle), \\
\sum_{a=0}^{d-1} \sum_{\lambda \in \Lambda} p(\lambda) p(a|1, \rho_A^\lambda) p(\mathbf{b}|1, \dots, 1, \lambda) &\leq \sum_{\lambda \in \Lambda} p(\lambda) \max_a p(a|1, \rho_A^\lambda) \leq \max_{|\psi\rangle \in \mathcal{C}^d} \max_a p(a|1, |\psi\rangle),
\end{aligned} \tag{B11}$$

where, to get the first inequalities in both cases, we used the fact that probabilities are normalized. To get the second inequality, we exploit the fact that Alice's probability distributions are quantum and hence linear functions of the state. Substituting this into our function, we get

$$\begin{aligned}
S_d(\boldsymbol{\alpha}) &\leq \max_{|\psi\rangle \in \mathcal{C}^d} \left[d(N-1) \max_a p(a|0, |\psi\rangle) + \gamma(\boldsymbol{\alpha}) \left(d \max_a p(a|1, |\psi\rangle) - \sum_{i=0}^{d-1} \alpha_i \sum_{a=0}^{d-1} \frac{p(a|0, |\psi\rangle)}{\alpha_a} \right) \right] \\
&\quad - (N-2),
\end{aligned} \tag{B12}$$

where we have used the fact that the maximization over the state $|\psi\rangle$ is happening over the whole expression. We can further write the state $|\psi\rangle$ in the computational basis as $|\psi\rangle = \sum_i \eta_i |i\rangle$ and then substitute the explicit forms of Alice's observables X and Z , which results in

$$\begin{aligned}
S_d(\boldsymbol{\alpha}) &\leq \max_{\substack{|\eta_0|, \dots, |\eta_{d-1}| \\ |\eta_0|^2 + \dots + |\eta_{d-1}|^2 = 1}} \left(d(N-1) \max_a |\eta_a|^2 + \gamma(\boldsymbol{\alpha}) \left[\left(\sum_{i=0}^{d-1} |\eta_i| \right)^2 - \sum_{i=0}^{d-1} \alpha_i \sum_{a=0}^{d-1} \frac{|\eta_a|^2}{\alpha_a} \right] \right) \\
&\quad - (N-2).
\end{aligned} \tag{B13}$$

If we can now show that the terms inside the square brackets are less than zero, we can conclude that $S_d(\boldsymbol{\alpha}) \leq (d-1)(N-1) + 1$. To do this, we notice the following:

$$\left(\sum_{i=0}^{d-1} |\eta_i| \right)^2 = \left(\sum_{i=0}^{d-1} \sqrt{\alpha_i} \frac{|\eta_i|}{\sqrt{\alpha_i}} \right)^2 \leq \sum_{i=0}^{d-1} \alpha_i \sum_{j=0}^{d-1} \frac{|\eta_j|^2}{\alpha_j}, \tag{B14}$$

where we have used the Cauchy-Schwarz inequality to get the last expression. Consequently, we have $S_d(\boldsymbol{\alpha}) \leq (d-1)(N-1) + 1$; however, from Eq. (B13), we see that $S_d(\boldsymbol{\alpha}) = (d-1)(N-1) + 1$ is possible iff $\max_a |\eta_a|^2 = 1$, and the terms inside the square brackets must also vanish. This is possible only if $\alpha_i = \lambda \eta_i$. Because $\sum_i \alpha_i^2 = \sum_i |\eta_i|^2 = 1$, we must have $\lambda = 1$; however, $\alpha_i < 1$ for all i , which does not allow for $\max_a |\eta_a|^2 = 1$ to hold. Therefore, $S_d(\boldsymbol{\alpha}) = (d-1)(N-1) + 1$ is never possible, which implies $S_d(\boldsymbol{\alpha}) < (d-1)(N-1) + 1$. ■

Let us now find the quantum bound of the steering functional $\langle S_d(\boldsymbol{\alpha}, N) \rangle$.

Fact 4. The quantum bound of the steering functional $\langle S_d(\boldsymbol{\alpha}, N) \rangle$, where $S_d(\boldsymbol{\alpha}, N)$ is given in (B2), is given by $\beta_Q = (d-1)(N-1) + 1$ and is achieved by the Schmidt

state $|\psi(\boldsymbol{\alpha})\rangle$ (B1) with the observables

$$B_{j,0} = Z, \quad B_{j,1} = X. \tag{B15}$$

Proof. Let us rewrite our functional as

$$S_d(\boldsymbol{\alpha}, N) = \sum_{k=1}^{d-1} \left(\sum_{j=1}^{N-1} A_0^k \otimes B_{j,0}^k \right) + T_d(\boldsymbol{\alpha}, N), \tag{B16}$$

with

$$T_d(\boldsymbol{\alpha}, N) = \sum_{k=1}^{d-1} \left(\gamma(\boldsymbol{\alpha}) A_1^k \otimes \bigotimes_{j=1}^{N-1} B_{j,1}^k + \delta_k(\boldsymbol{\alpha}) A_0^k \right). \tag{B17}$$

Since A_0 is unitary and $(B_{j,0}^k)^\dagger B_{j,0}^k \leq \mathbb{1}$ for any k and j , we have $|\langle \psi_N | A_0^k \otimes B_{j,0}^k | \psi_N \rangle| \leq 1$ for all $k \in \{1, \dots, d-1\}$, $j \in \{1, \dots, N-1\}$, and $|\psi_N\rangle$. This implies that

$$\begin{aligned}
&\max_{|\psi_N\rangle} \langle \psi_N | S_d(\boldsymbol{\alpha}, N) | \psi_N \rangle \\
&\leq (d-1)(N-1) + \max_{|\psi_N\rangle} \langle \psi_N | T_d(\boldsymbol{\alpha}, N) | \psi_N \rangle.
\end{aligned} \tag{B18}$$

Therefore, we only need to prove that $\langle \psi_N | T_d(\boldsymbol{\alpha}, N) | \psi_N \rangle \leq 1$. We can represent any pure state in the Hilbert space

$\mathcal{H}_A \otimes_{i=1}^{N-1} \mathcal{H}_{B_i}$ as

$$|\psi_N\rangle_{AB} = \sum_{i=0}^{d-1} \lambda_i |i\rangle_A |e_i\rangle_B, \quad (\text{B19})$$

where $\lambda_i \geq 0$ and $\lambda_0^2 + \dots + \lambda_{d-1}^2 = 1$, and $|e_i\rangle$ are some vectors in the Hilbert space $\bigotimes_{i=1}^{N-1} \mathcal{H}_{B_i}$ such that $|e_i\rangle$ are not necessarily orthogonal. With this representation, we can rewrite $\langle\psi_N|T_d(\alpha, N)|\psi_N\rangle$ as

$$\begin{aligned} \langle\psi_N|T_d(\alpha, N)|\psi_N\rangle &= \sum_{k=1}^{d-1} \sum_{i,j=0}^{d-1} \left[\gamma(\alpha) \lambda_i \lambda_j \langle i|A_1^k|j\rangle \langle e_i| \bigotimes_{j=1}^{N-1} B_{j,1}^k |e_j\rangle + \delta_k(\alpha) \lambda_i \lambda_j \langle i|A_0^k|j\rangle \langle e_i|e_j\rangle \right] \\ &= \sum_{k=1}^{d-1} \sum_{i=0}^{d-1} \left[\gamma(\alpha) \lambda_i \lambda_{i-k} \langle e_i| \bigotimes_{j=1}^{N-1} B_{j,1}^k |e_j\rangle + \delta_k(\alpha) \lambda_i^2 \omega^{ik} \right], \end{aligned} \quad (\text{B20})$$

where for the second equality, we have substituted A_i from (1). Next, we note that

$$\begin{aligned} \sum_{k=1}^{d-1} \sum_{i=0}^{d-1} \delta_k(\alpha) \lambda_i^2 \omega^{ik} &= 1 + \sum_{k=0}^{d-1} \sum_{i=0}^{d-1} \delta_k(\alpha) \lambda_i^2 \omega^{ik} \\ &= 1 + \gamma(\alpha) - \gamma(\alpha) \sum_{i,j=0}^{d-1} \frac{\alpha_i}{\alpha_j} \lambda_j^2, \end{aligned} \quad (\text{B21})$$

where we get the first equality by using $\delta_0(\alpha) = -1$ and the second equality by using the explicit form of $\delta_k(\alpha)$. By substituting the above expression into Eq. (B20), we get

$$\begin{aligned} \langle\psi_N|T_d(\alpha, N)|\psi_N\rangle &= 1 + \gamma(\alpha) \sum_{k=0}^{d-1} \sum_{i=0}^{d-1} \lambda_i \lambda_{i-k} \text{Re} \left(\langle e_i| \bigotimes_{j=1}^{N-1} B_{j,1}^k |e_j\rangle \right) \\ &\quad - \gamma(\alpha) \sum_{i,j=0}^{d-1} \frac{\alpha_i}{\alpha_j} \lambda_j^2, \end{aligned} \quad (\text{B22})$$

since $\mathcal{S}_d$ is a Hermitian operator and every λ_i is positive. In this expression, we use the fact that $\text{Re}(z) \leq |z|$ and $(B_{j|0}^k)^\dagger B_{j|0}^k = \mathbb{1}$ for any k and j to obtain

$$\begin{aligned} \langle\psi_N|T_d(\alpha, N)|\psi_N\rangle &\leq 1 + \gamma(\alpha) \left[\left(\sum_{i=0}^{d-1} \lambda_i^2 \right)^2 - \sum_{i=0}^{d-1} \alpha_i \sum_{j=0}^{d-1} \frac{\lambda_j^2}{\alpha_j} \right]. \end{aligned} \quad (\text{B23})$$

We see that the term in square brackets has the same form as the terms in square brackets of Eq. (B13), which is

upper bounded by 0. This implies that

$$\begin{aligned} \langle\psi_N|T_d(\alpha, N)|\psi_N\rangle &\leq 1 \\ \implies \mathcal{S}_d(\alpha, N) &\leq (d-1)(N-1) + 1. \end{aligned} \quad (\text{B24})$$

It is now simple to observe that with the Schmidt state $|\psi(\alpha)\rangle$ from (B1) and observables

$$B_{j,0} = Z, \quad B_{j,1} = X, \quad (\text{B25})$$

one can attain the value $\langle\mathcal{S}_d(\alpha, N)\rangle = (d-1)(N-1) + 1$. This completes the proof. ■

Consequently, if all the parties attain the quantum bound $(d-1)(N-1) + 1$ of $\mathcal{S}_d(\alpha, N)$, the following conditions hold true:

$$\langle\psi_N|A_0^k \otimes B_{j,0}^k |\psi_N\rangle = 1, \quad \langle\psi_N|T_d(\alpha, N)|\psi_N\rangle = 1, \quad (\text{B26})$$

for every $j \in \{1, \dots, N-1\}$ and every $k \in \{1, \dots, d-1\}$, where $T_d(\alpha, N)$ is given in (B17). These conditions hold because of Eqs. (B16), (B18), and (B24). As a result, we have the following conditions for the state $|\psi_N\rangle$ and observables A_i and $B_{j,i}$:

$$A_0^k \otimes B_{j,0}^k |\psi_N\rangle = |\psi_N\rangle, \quad (\text{B27})$$

$$\sum_{k=1}^{d-1} \left(\gamma(\alpha) A_{j,1}^{k,N-1} \otimes B_{j,1}^k + \delta_k(\alpha) A_0^k \right) |\psi_N\rangle = |\psi_N\rangle \quad (\text{B28})$$

for every $j \in \{1, \dots, N-1\}$ and every $k \in \{1, \dots, d-1\}$. Since A_i 's and B_i 's are unitary observables, we can rewrite Eq. (B27) as

$$(A_0^{(-k)} \otimes \mathbb{1}) |\psi_N\rangle = (\mathbb{1} \otimes B_{j,0}^{(k)}) |\psi_N\rangle. \quad (\text{B29})$$

The measurements chosen by Alice are $A_0 = Z$ and $A_1 = X$. Thus, we have the following relation among Alice's

observables:

$$A_1 A_0 = \omega A_0 A_1, \quad (\text{B30})$$

where $\omega = e^{\frac{2\pi i}{d}}$. As we are about to present, the above relations will be particularly useful for self-testing the Schmidt states.

Theorem B1. Assume that the steering inequality (B5) is maximally violated when the 0th party, Alice, chooses the observables A_i , defined to be $A_0 = Z$ and $A_1 = X$, and all the other parties choose their observables $B_{j,i}$ for $i \in \{0, 1\}$ and $j = 1, 2, \dots, N-1$ acting on $\mathcal{H}_{B_j}$. Let us say that the state attaining this violation is given by $|\psi_N\rangle_{ABE} \in \mathbb{C}^d \otimes \bigotimes_{i=1}^{N-1} \mathcal{H}_{B_i} \otimes \mathcal{H}_E$; the following statement then holds true for any d : $\mathcal{H}_{B_i} = \mathcal{H}_{B'_i} \otimes \mathcal{H}_{B''_i}$, where $\mathcal{H}_{B'_i} \equiv \mathbb{C}^d$, $\mathcal{H}_{B''_i}$ is some finite-dimensional Hilbert space, and there exists a local unitary transformation on Bob's side $U_j : \mathcal{H}_{B_j} \rightarrow \mathcal{H}_{B_j}$, such that

$$\forall j, \quad U_j B_{j,0} U_j^\dagger = Z_{B'_j} \otimes \mathbb{1}_{B''_j}, \quad U_j B_{j,1} U_j^\dagger = X_{B'_j} \otimes \mathbb{1}_{B''_j}, \quad (\text{B31})$$

where B'' denotes Bob's auxiliary system, and the state $|\psi_N\rangle$ is given by

$$\left(\mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_i \right) |\psi_N\rangle_{ABE} = |\psi(\alpha)\rangle_{AB'} \otimes |\xi\rangle_{B''E}, \quad (\text{B32})$$

where $|\xi\rangle_{B''E} \in \bigotimes_{j=1}^{N-1} \mathcal{H}_{B''_j} \otimes \mathcal{H}_E$ denotes the auxiliary state.

Proof. We begin by taking (B28) and multiplying it by $B_{l,1} \otimes \mathbb{1}$ on both sides:

$$\begin{aligned} & \left(\sum_{k=1}^{d-1} \gamma(\alpha) A_1^k \otimes B_{l,1}^{k+1} \bigotimes_{j=1}^{N-1} B_{j,1}^k \right) |\psi_N\rangle \\ &= \left(\mathbb{1} - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) \otimes B_{l,1} |\psi_N\rangle. \end{aligned} \quad (\text{B33})$$

Now, we multiply the above by $B_{l,0} \otimes \mathbb{1}$ on both the sides:

$$\begin{aligned} & \left(\sum_{k=1}^{d-1} \delta_k(\alpha) A_1^k \otimes B_{l,0} B_{l,1}^{k+1} \bigotimes_{j=1}^{N-1} B_{j,1}^k \right) |\psi_N\rangle \\ &= \left(\mathbb{1} - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) \otimes B_{l,0} B_{l,1} |\psi_N\rangle. \end{aligned} \quad (\text{B34})$$

Similarly, we multiply both the sides of (B33) by $A_0^{-1} \otimes \mathbb{1}$ to obtain

$$\begin{aligned} & \left(\sum_{k=1}^{d-1} \gamma(\alpha) A_0^{-1} A_1^k \otimes B_{l,1}^{k+1} \bigotimes_{j=1}^{N-1} B_{j,1}^k \right) |\psi_N\rangle \\ &= \left(\mathbb{1} - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) A_0^{-1} \otimes B_{l,1} |\psi_N\rangle. \end{aligned} \quad (\text{B35})$$

By using the commutation relation (B30), and since $B_{j,1} \otimes \mathbb{1}$ and $A_0 \otimes \mathbb{1}$ commute, we get

$$\begin{aligned} & \left(\sum_{k=1}^{d-1} \gamma(\alpha) \omega^k A_1^k \otimes B_{l,1}^{k+1} \bigotimes_{j=1}^{N-1} B_{j,1}^k \right) A_0^{-1} |\psi_N\rangle \\ &= \left(\mathbb{1} - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) A_0^{-1} \otimes B_{l,1} |\psi_N\rangle. \end{aligned} \quad (\text{B36})$$

In the above equation, we use (B29) for $j = l$, which gives

$$\begin{aligned} & \left(\sum_{k=1}^{d-1} \gamma(\alpha) \omega^k A_1^k \otimes B_{l,1}^{k+1} B_{l,0} \bigotimes_{j=1}^{N-1} B_{j,1}^k \right) |\psi_N\rangle \\ &= \left(\mathbb{1} - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) \otimes B_{l,1} B_{l,0} |\psi_N\rangle. \end{aligned} \quad (\text{B37})$$

We can proceed now by multiplying (B37) by ω and subtracting it from (B34) as follows:

$$\begin{aligned} & \left(\sum_{k=1}^{d-1} \gamma(\alpha) A_1^k \otimes (B_{l,0} B_{l,1}^{k+1} - \omega^{k+1} B_{l,1}^{k+1} B_{l,0}) \bigotimes_{j=1}^{N-1} B_{j,1}^k \right) |\psi_N\rangle \\ &= \left(\mathbb{1} - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) \otimes (B_{l,0} B_{l,1} - \omega B_{l,1} B_{l,0}) |\psi_N\rangle. \end{aligned} \quad (\text{B38})$$

Next, we again take (B28) and multiply it by $A_1^{-1} \otimes \mathbb{1}$ on both sides:

$$\begin{aligned} & \left(\sum_{k=1}^{d-1} \gamma(\alpha) A_1^{k-1} \otimes B_{l,1}^k \bigotimes_{j=1}^{N-1} B_{j,1}^k - \delta_k(\alpha) A_1^{-1} A_0^k \right) |\psi_N\rangle \\ &= A_1^{-1} |\psi_N\rangle. \end{aligned} \quad (\text{B39})$$

We multiply the above by $B_{l,0} \otimes \mathbb{1}$, use the fact that $B_{l,0} \otimes \mathbb{1}$ commutes with $A_1 \otimes \mathbb{1}$, and then use (B29) for $k = 1$ to

obtain

$$\left(\sum_{k=1}^{d-1} \gamma(\alpha) A_1^{k-1} \otimes B_{l,0} B_{l,1}^k \otimes_{j=1}^{N-1} B_{j,1}^k - \delta_k(\alpha) A_1^{-1} A_0^{k-1} \right) |\psi_N\rangle = A_1^{-1} A_0^{-1} |\psi_N\rangle. \quad (\text{B40})$$

Now, we multiply both sides of (B39) by $A_0^{-1} \otimes \mathbb{1}$:

$$\left(\sum_{k=1}^{d-1} \gamma(\alpha) A_0^{-1} A_1^{k-1} \otimes B_{l,1}^k \otimes_{j=1}^{N-1} B_{j,1}^k - \delta_k(\alpha) A_0^{-1} A_1^{-1} A_0^k \right) |\psi_N\rangle = A_0^{-1} A_1^{-1} |\psi_N\rangle. \quad (\text{B41})$$

Using the commutation relation (B30), the condition (B29), and the fact that $B_{j,1} \otimes \mathbb{1}$ and $A_0 \otimes \mathbb{1}$ commute, we obtain

$$\left(\sum_{k=1}^{d-1} \gamma(\alpha) \omega^{k-1} A_1^{k-1} \otimes B_{l,1}^k B_{l,0} \otimes_{j=1}^{N-1} B_{j,1}^k - \omega^{-1} \delta_k(\alpha) A_1^{-1} A_0^{k-1} \right) \times |\psi_N\rangle = \omega^{-1} A_1^{-1} A_0^{-1} |\psi_N\rangle. \quad (\text{B42})$$

We proceed now by multiplying (B40) by ω and subtracting it from (B42) to obtain

$$\left(\sum_{k=1}^{d-1} \gamma(\alpha) A_1^{k-1} \otimes (B_{l,0} B_{l,1}^k - \omega^k B_{l,1}^k B_{l,0}) \otimes_{j=1}^{N-1} B_{j,1}^k \right) |\psi_N\rangle = 0, \quad (\text{B43})$$

which can be broken up as

$$\left(\sum_{k=2}^{d-1} \gamma(\alpha) A_1^{k-1} \otimes (B_{l,0} B_{l,1}^k - \omega^k B_{l,1}^k B_{l,0}) \otimes_{j=1}^{N-1} B_{j,1}^k \right) |\psi_N\rangle = -\gamma(\alpha) \left((B_{l,0} B_{l,1} - \omega B_{l,1} B_{l,0}) \otimes_{j=1}^{N-1} B_{j,1} \right) |\psi_N\rangle. \quad (\text{B44})$$

The above can be further simplified as

$$\left(\sum_{k=2}^{d-1} \gamma(\alpha) A_1^{k-1} \otimes (B_{l,0} B_{l,1}^k - \omega^k B_{l,1}^k B_{l,0}) \otimes_{j=1}^{N-1} B_{j,1}^{k-1} \right) |\psi_N\rangle = -\gamma(\alpha) (B_{l,0} B_{l,1} - \omega B_{l,1} B_{l,0}) |\psi_N\rangle. \quad (\text{B45})$$

Now, we can see that the left-hand sides of (B38) and (B45) are the same. Thus, we can conclude that

$$\left(\mathbb{1} - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) \otimes (B_{l,0} B_{l,1} - \omega B_{l,1} B_{l,0}) |\psi_N\rangle = -\gamma(\alpha) (B_{l,0} B_{l,1} - \omega B_{l,1} B_{l,0}) |\psi_N\rangle. \quad (\text{B46})$$

We finally get

$$\left(\mathbb{1} (1 + \gamma(\alpha)) - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) \otimes (B_{l,0} B_{l,1} - \omega B_{l,1} B_{l,0}) |\psi_N\rangle = 0. \quad (\text{B47})$$

It can be shown that if $(P \otimes Q)v = 0$ for any $v \neq 0$ and P is invertible, then we can conclude that $Q = 0$. We can use this fact to show that $\left(\mathbb{1} (1 + \gamma(\alpha)) - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right)$ is invertible. Indeed, by expanding it term by term, we get

$$\left(\mathbb{1} (1 + \gamma(\alpha)) - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k \right) = \sum_{l=0}^{d-1} \left((1 + \gamma(\alpha)) - \sum_{k=1}^{d-1} \delta_k(\alpha) \omega^{kl} \right) |l\rangle \langle l|. \quad (\text{B48})$$

Notice that the above matrix is not invertible if any of the elements is 0. Thus, we solve the right-hand side of the above equation for the l th element:

$$\begin{aligned} (1 + \gamma(\alpha)) + \frac{\gamma(\alpha)}{d} \left(\sum_{k=1}^{d-1} \sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j} \omega^{k(l-j)} \right) \\ = (1 + \gamma(\alpha)) + \frac{\gamma(\alpha)}{d} \left(\sum_{\substack{i=0 \\ i \neq l}}^{d-1} \frac{\alpha_i}{\alpha_l} (d-1) - \sum_{\substack{i,j=0 \\ i \neq j, j \neq l}}^{d-1} \frac{\alpha_i}{\alpha_j} \right) \\ = (1 + \gamma(\alpha)) + \gamma(\alpha) \left(\sum_{i=0}^{d-1} \frac{\alpha_i}{\alpha_l} - \frac{1}{d} \sum_{i,j=0}^{d-1} \frac{\alpha_i}{\alpha_j} \right), \end{aligned} \quad (\text{B49})$$

where we have used the fact that $\sum_{k=0}^{d-1} \omega^{k(l-j)} = d\delta_{l,j}$. Substituting $\gamma(\alpha)$ from (B3), we get

$$\begin{aligned} (1 + \gamma(\alpha)) + \gamma(\alpha) \left(\sum_{i=0}^{d-1} \frac{\alpha_i}{\alpha_l} - \frac{1}{d} \sum_{i,j=0}^{d-1} \frac{\alpha_i}{\alpha_j} \right) \\ = \gamma(\alpha) \left(1 + \frac{1}{d} \sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j} - \frac{1}{d} \sum_{i,j=0}^{d-1} \frac{\alpha_i}{\alpha_j} + \sum_{i=0}^{d-1} \frac{\alpha_i}{\alpha_l} \right), \\ = \gamma(\alpha) \left(\frac{1}{d} \sum_{i,j=0}^{d-1} \frac{\alpha_i}{\alpha_j} - \frac{1}{d} \sum_{i,j=0}^{d-1} \frac{\alpha_i}{\alpha_j} + \sum_{i=0}^{d-1} \frac{\alpha_i}{\alpha_l} \right) \\ = \gamma(\alpha) \left(\sum_{i=0}^{d-1} \frac{\alpha_i}{\alpha_l} \right) > 0. \end{aligned} \quad (\text{B50})$$

Thus, we can conclude that the matrix $\left(\mathbb{1}(1 + \gamma(\alpha)) - \sum_{k=1}^{d-1} \delta_k(\alpha) A_0^k\right)$ is invertible. Therefore, we have

$$\mathbb{1} \otimes (B_{l,0} B_{l,1} - \omega B_{l,1} B_{l,0}) |\psi_N\rangle = 0. \quad (\text{B51})$$

Finally, we can conclude that $B_{l,0} B_{l,1} = \omega B_{l,1} B_{l,0}$. As was shown in Ref. [13], if two observables B_0 and B_1 follow the above relation, then there exists a unitary matrix U such that

$$UB_{l,0}U^\dagger = Z_{B'_l}^\dagger \otimes \mathbb{1}_{B''_l}, \quad UB_{l,1}U^\dagger = X_{B'_l} \otimes \mathbb{1}_{B''_l}. \quad (\text{B52})$$

From the above relations, we can use (B27) and (B28) to show that the state saturating the quantum bound is the Schmidt state of local dimension d .

Now, let us prove that the state maximally violating the steering inequality must be equivalent to the Schmidt state, up to local unitary transformations. By taking a general state of the form $|\psi_N\rangle = \sum_{i,j_1,j_2,\dots,j_{N-1}=0}^{d-1} |ij_1j_2\dots j_{N-1}\rangle |\psi_{ij_1j_2\dots j_{N-1}}\rangle$, we can plug it into (B27) with $k = 1$ for the l th Bob to obtain

$$\begin{aligned} (Z \otimes Z_l^\dagger \otimes \mathbb{1}) \left(\sum_{i,j_1,j_2,\dots,j_{N-1}=0}^{d-1} |ij_1j_2\dots j_{N-1}\rangle |\psi_{ij_1j_2\dots j_{N-1}}\rangle \right) \\ = \sum_{i,j_1,j_2,\dots,j_{N-1}=0}^{d-1} |ij_1j_2\dots j_{N-1}\rangle |\psi_{ij_1j_2\dots j_{N-1}}\rangle, \end{aligned} \quad (\text{B53})$$

which is equivalent to

$$\begin{aligned} \sum_{i,j_1,j_2,\dots,j_{N-1}=0}^{d-1} \omega^{i-j_l} |ij_1j_2\dots j_{N-1}\rangle |\psi_{ij_1j_2\dots j_{N-1}}\rangle \\ = \sum_{i,j_1,j_2,\dots,j_{N-1}=0}^{d-1} |ij_1j_2\dots j_{N-1}\rangle |\psi_{ij_1j_2\dots j_{N-1}}\rangle. \end{aligned} \quad (\text{B54})$$

For $i \neq j_l$, the above identity implies $\omega^{j_l-i} |\psi_{ij_1j_2\dots j_{N-1}}\rangle = |\psi_{ij_1j_2\dots j_{N-1}}\rangle$. This indicates that

$$\forall i, j_l, \text{ such that } i \neq j_l, \quad |\psi_{ij_1j_2\dots j_{N-1}}\rangle = 0. \quad (\text{B55})$$

Consequently, we must have $i = j_l$ for all l . Next, we can proceed similarly with Eq. (B28) to obtain

$$\begin{aligned} \left(\sum_{k=1}^{d-1} \gamma(\alpha) X_{\otimes_{j=1}^{N-1}}^k \otimes \mathbb{1} + \delta_k(\alpha) Z^k \right) \left(\sum_{i=0}^{d-1} |ii\dots i\rangle |\psi_{ii\dots i}\rangle \right) \\ = \sum_{i=0}^{d-1} |ii\dots i\rangle |\psi_{ii\dots i}\rangle. \end{aligned} \quad (\text{B56})$$

After performing the operations, we get

$$\begin{aligned} \sum_{i=0}^{d-1} \left(\sum_{k=1}^{d-1} \gamma(\alpha) |i+k\rangle |i+k\rangle \dots |i+k\rangle + \omega^{ki} \delta_k(\alpha) |ii\dots i\rangle \right) \\ \times |\psi_{ii\dots i}\rangle = \sum_{i=0}^{d-1} |ii\dots i\rangle |\psi_{ii\dots i}\rangle. \end{aligned} \quad (\text{B57})$$

If we take the inner product of the above relation with $\langle ss\dots s|$, we obtain

$$\sum_{k=1}^{d-1} (\gamma(\alpha) |\psi_{s-k,s-k,\dots,s-k}\rangle + \omega^{ks} \delta_k(\alpha)) |\psi_{ss\dots s}\rangle = |\psi_{ss\dots s}\rangle. \quad (\text{B58})$$

From (B3), we have

$$\begin{aligned} \sum_{k=1}^{d-1} \left(|\psi_{s-k,s-k,\dots,s-k}\rangle - \frac{\omega^{ks}}{d} \sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j} \omega^{k(d-j)} \right) |\psi_{ss\dots s}\rangle \\ = \frac{1}{d} \sum_{\substack{i,j=0 \\ i \neq j}}^{d-1} \frac{\alpha_i}{\alpha_j} |\psi_{ss\dots s}\rangle, \end{aligned} \quad (\text{B59})$$

which can be simplified in the following way:

$$\sum_{k=1}^{d-1} |\psi_{s-k,s-k,\dots,s-k}\rangle = \frac{\sum_{i=0}^{d-1} \alpha_i}{\alpha_s} |\psi_{ss\dots s}\rangle. \quad (\text{B60})$$

Thus, for any s , we have

$$|\psi_{ss\dots s}\rangle = \frac{\sum_{k=1}^{d-1} |\psi_{s-k,s-k,\dots,s-k}\rangle}{\sum_{i=0}^{d-1} \alpha_i} \alpha_s, \quad \forall s \in \{0, 1, \dots, d-1\}. \quad (\text{B61})$$

If we define

$$|\xi\rangle_{\mathbf{B}'E} := \frac{\sum_{k=1}^{d-1} |\psi_{s-k,s-k,\dots,s-k}\rangle}{\sum_{i=0}^{d-1} \alpha_i}, \quad (\text{B62})$$

then we can write

$$\left(\mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_j \right) |\psi_N\rangle_{ABE} = \left(\sum_{i=0}^{d-1} \alpha_i |i\rangle^{\otimes N} \right) \otimes |\xi\rangle_{\mathbf{B}'E}, \quad (\text{B63})$$

which completes the proof. $\blacksquare$

APPENDIX C: SELF-TEST N -QUBIT GENERALIZED W STATES

The N -qubit generalized W states have the following form:

$$\begin{aligned} |\psi_{W_N}\rangle &= \sum_{l=0}^{N-1} \alpha_{l+1} |0 \dots, 0, 1_l, 0, \dots, 0\rangle \\ &= \alpha_1 |10 \dots 00\rangle + \alpha_2 |01 \dots 00\rangle + \dots \\ &\quad + \alpha_N |00 \dots 01\rangle, \end{aligned} \quad (C1)$$

such that $\alpha_i > 0$ for every i and $\sum_{i=1}^N \alpha_i^2 = 1$. We introduce a steering inequality that is maximally violated by the above state through the following steering operator:

$$\begin{aligned} \mathcal{W}_N(\alpha) &= -2Z_A \otimes \bigotimes_{k=1}^{N-1} B_{k,0} + \sum_{l=1}^{N-1} [Z_A \otimes \mathbb{1}_l \otimes (\mathbb{1} - P_l) \\ &\quad + (\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l) \otimes P_l] \\ &\quad + \sum_{l=1}^{N-1} [\mathbb{1}_A \otimes B_{l,0} \otimes (\mathbb{1} - P_l) + \mathbb{1}_A \otimes \mathbb{1}_l \otimes P_l], \end{aligned} \quad (C2)$$

where

$$\gamma_l = \frac{2\alpha_{l+1}\alpha_1}{\alpha_{l+1}^2 + \alpha_1^2}, \quad \delta_l = \frac{\alpha_{l+1}^2 - \alpha_1^2}{\alpha_{l+1}^2 + \alpha_1^2}, \quad \forall l \in \{1, \dots, N-1\}, \quad (C3)$$

with

$$P_l = \frac{1}{2^{N-2}} \bigotimes_{k=1, k \neq l}^{N-1} (\mathbb{1}_k + B_{k,0}), \quad (C4)$$

and $B_{k,0}$ and $B_{k,1}$ are the measurements acting on the k th qubit. In the same way, $\mathbb{1}_l$ is the identity operation acting on the l th qubit. The corresponding steering inequality is given by

$$\langle \mathcal{W}_N(\alpha) \rangle \leq \beta_L, \quad (C5)$$

where β_L is the LHS bound.

Let us now find the quantum bound of the steering functional $\langle \mathcal{W}_N(\alpha) \rangle$.

Fact 1. The quantum bound of the steering functional $\langle \mathcal{W}_N(\alpha) \rangle$, where $\mathcal{W}_N(\alpha)$ is given in (C2), is $\beta_Q = 2N$ and is achieved by the N -qubit generalized W state $|\psi_{W_N}\rangle$ (C1) with the observables

$$B_{j,0} = Z, \quad B_{j,1} = X. \quad (C6)$$

Proof. Before proceeding, let us recall here that $B_{j,0}$ are unitary observables, as they correspond to projective measurements. Now, let us observe that the first term in the steering operator $\mathcal{W}_N$ (C2) is

$$Z_A \otimes \bigotimes_{k=1}^{N-1} B_{k,0} \leq \mathbb{1}, \quad (C7)$$

where we used the fact that $|B_{k,0}| \leq \mathbb{1}$. Similarly, one can straightforwardly conclude that for any l ,

$$\mathbb{1}_A \otimes B_{l,0} \otimes (\mathbb{1} - P_l) + \mathbb{1}_A \otimes \mathbb{1}_l \otimes P_l \leq \mathbb{1}. \quad (C8)$$

Let us now observe that for any l , $B_{l,1} = \Pi_l^+ - \Pi_l^-$, where Π_l^+ and Π_l^- denote the projector onto the positive and negative eigenspace, respectively. Now, we observe that

$$\begin{aligned} &|\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l| \\ &\leq |(\gamma_l X_A + \delta_l Z_A) \otimes \Pi_l^+| + |(-\gamma_l X_A + \delta_l Z_A) \otimes \Pi_l^-|. \end{aligned} \quad (C9)$$

As Π_l^+ and Π_l^- are positive matrices, we can conclude from the above formula that

$$\begin{aligned} &|\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l| \\ &\leq \lambda_{\max}^{(1)} \mathbb{1}_A \otimes \Pi_l^+ + \lambda_{\max}^{(2)} \mathbb{1}_A \otimes \Pi_l^-, \end{aligned} \quad (C10)$$

where $\lambda_{\max}^{(1)}$ and $\lambda_{\max}^{(2)}$ are the maximum eigenvalues of $|\gamma_l X_A + \delta_l Z_A|$ and $|-\gamma_l X_A + \delta_l Z_A|$, respectively. One can immediately evaluate that $\lambda_{\max}^{(1)} = \lambda_{\max}^{(2)} = 1$, and thus we obtain

$$|\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l| \leq \mathbb{1}_{A,l}. \quad (C11)$$

Finally, considering the second term in (C2) for any l , we have

$$\begin{aligned} &Z_A \otimes \mathbb{1}_l \otimes (\mathbb{1} - P_l) + (\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l) \otimes P_l \\ &\leq \mathbb{1}_{A,l} \otimes (\mathbb{1} - P_l) + |\gamma_l X_A \otimes B_{l,1} \\ &\quad + \delta_l Z_A \otimes \mathbb{1}_l| \otimes P_l \leq \mathbb{1}. \end{aligned} \quad (C12)$$

Thus, the maximum quantum value of the steering inequality (C2) is $2N$, and this can be achieved when $B_{j,0} = Z$ and $B_{j,1} = X$ with the generalized W state $|\psi_{W_N}\rangle$. ■

To achieve the maximal violation of a steering inequality through operator (C2), the following conditions need to

be satisfied:

$$Z_A \otimes \bigotimes_{k=1}^{N-1} B_{k,0} |\psi\rangle_{AB} = -|\psi\rangle_{AB}, \quad (\text{C13})$$

and for all $l = 1, \dots, N-1$,

$$\begin{aligned} & [Z_A \otimes \mathbb{1}_l (\mathbb{1} - P_l) + (\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l) \\ & \otimes P_l] |\psi\rangle_{AB} = |\psi\rangle_{AB} \end{aligned} \quad (\text{C14})$$

and

$$[\mathbb{1}_A \otimes B_{l,0} \otimes (\mathbb{1} - P_l) + \mathbb{1}_A \otimes \mathbb{1}_l \otimes P_l] |\psi\rangle_{AB} = |\psi\rangle_{AB}. \quad (\text{C15})$$

Now, let us proceed to the self-testing statement.

Theorem C1. Assume that the steering inequality (C5) is maximally violated when the 0th party, Alice, chooses the observables A_i defined to be $A_0 = Z$ and $A_1 = X$, and all the other parties choose their observables as $B_{j,i}$ for $i \in \{0, 1\}$ and $j = 1, 2, \dots, N-1$ acting on $\mathcal{H}_{B_j}$. Let us say that the state attaining this violation is given by $|\psi_N\rangle_{ABE} \in \mathbb{C}^2 \otimes \bigotimes_{j=1}^{N-1} \mathcal{H}_{B_j} \otimes \mathcal{H}_E$; then, the following statement holds true: $\mathcal{H}_{B_j} = \mathcal{H}_{B'_j} \otimes \mathcal{H}_{B''_j}$, where $\mathcal{H}_{B'_j} \equiv \mathbb{C}^2$, $\mathcal{H}_{B''_j}$ is some finite-dimensional Hilbert space, and there exists a local unitary transformation on Bob's side $U_j : \mathcal{H}_{B_j} \rightarrow \mathcal{H}_{B'_j}$, such that

$$\forall j, \quad U_j B_{j,0} U_j^\dagger = Z_{B'_j} \otimes \mathbb{1}_{B''_j}, \quad U_j B_{j,1} U_j^\dagger = X_{B'_j} \otimes \mathbb{1}_{B''_j}, \quad (\text{C16})$$

where B'' denotes Bob's auxiliary system and the state $|\psi_N\rangle$ is given by

$$\left(\mathbb{1}_{AE} \otimes \bigotimes_{i=1}^{N-1} U_i \right) |\psi_N\rangle_{ABE} = |\psi_{W_N}\rangle_{AB'} \otimes |\xi\rangle_{B''E}, \quad (\text{C17})$$

where $|\xi\rangle_{B''E} \in \bigotimes_{j=1}^{N-1} \mathcal{H}_{B''_j} \otimes \mathcal{H}_E$ denotes the auxiliary state.

Proof. For simplicity, we drop the lower indices from the state $|\psi\rangle_{ABE}$. First, let us consider (C13), rewrite it as

$$Z_A \otimes B_{l,0} \otimes \bigotimes_{k=1, k \neq l}^{N-1} B_{k,0} |\psi\rangle = -|\psi\rangle, \quad (\text{C18})$$

multiply it by $B_{l',0}$ for $l \neq l'$, and use the fact that $B_{l',0}^2 = \mathbb{1}$. By adding the result to (C13), we obtain

$$\begin{aligned} & Z_A \otimes B_{l,0} \otimes (\mathbb{1} + B_{l',0}) \otimes \bigotimes_{k=1, k \neq l, l'}^{N-1} B_{k,0} |\psi\rangle \\ & = -\mathbb{1} \otimes (\mathbb{1} + B_{l',0}) |\psi\rangle. \end{aligned} \quad (\text{C19})$$

Then, multiplying (C19) by $B_{l',0}$ and then adding the resulting formula to (C19), we obtain

$$\begin{aligned} & Z_A \otimes B_{l,0} \otimes (\mathbb{1} + B_{l',0}) \otimes (\mathbb{1} + B_{l'',0}) \\ & \otimes \bigotimes_{k=1, k \neq l, l', l''}^{N-1} B_{k,0} |\psi\rangle = -\mathbb{1} \otimes (\mathbb{1} + B_{l',0}) \otimes (\mathbb{1} + B_{l'',0}) |\psi\rangle. \end{aligned} \quad (\text{C20})$$

Continuing in a similar manner, we get

$$Z_A \otimes B_{l,0} \otimes P_l |\psi\rangle = -\mathbb{1} \otimes P_l |\psi\rangle, \quad (\text{C21})$$

where P_l is given in Eq. (C4). Denoting $|\tilde{\psi}\rangle := P_l |\psi\rangle$, we have

$$Z_A \otimes B_{l,0} |\tilde{\psi}\rangle = -|\tilde{\psi}\rangle. \quad (\text{C22})$$

Next, we consider (C14) for any l and multiply it by P_l , which, using the fact that $P_l^2 = P_l$, gives us

$$(\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l) |\tilde{\psi}\rangle = |\tilde{\psi}\rangle, \quad (\text{C23})$$

which, using the fact that $B_{l,1}$ is unitary, can be rearranged to

$$\gamma_l X_A |\tilde{\psi}\rangle = (\mathbb{1} - \delta_l Z_A \otimes \mathbb{1}_l) \otimes B_{l,1} |\tilde{\psi}\rangle. \quad (\text{C24})$$

Now, let us multiply the above by $Z_A \otimes \mathbb{1}_l$ and use the fact that this term commutes with $(\mathbb{1}_A - \delta_l Z_A) \otimes B_{l,1}$. That gives us

$$\gamma_l Z_A X_A \otimes \mathbb{1}_l |\tilde{\psi}\rangle = [(\mathbb{1}_A - \delta_l Z_A) \otimes B_{l,1}] (Z_A \otimes \mathbb{1}_l) |\tilde{\psi}\rangle, \quad (\text{C25})$$

which, using (C22) gives us

$$\gamma_l Z_A X_A \otimes \mathbb{1}_l |\tilde{\psi}\rangle = -(\mathbb{1}_A - \delta_l Z_A) \otimes B_{l,1} B_{l,0} |\tilde{\psi}\rangle. \quad (\text{C26})$$

Now, let us consider (C22) and multiply it by $\gamma_l X_A \otimes \mathbb{1}_l$. Similarly, we obtain

$$\gamma_l X_A Z_A \otimes \mathbb{1}_l |\tilde{\psi}\rangle = -(\mathbb{1}_A \otimes B_{l,0}) (\gamma_l X_A \otimes \mathbb{1}_l) |\tilde{\psi}\rangle. \quad (\text{C27})$$

Using (C24), the above formula can be expressed as

$$\gamma_l X_A Z_A \otimes \mathbb{1}_l |\tilde{\psi}\rangle = -(\mathbb{1}_A - \delta_l Z_A) \otimes B_{l,0} B_{l,1} |\tilde{\psi}\rangle. \quad (\text{C28})$$

By adding (C28) to (C26) and taking into account that $ZX + XZ = 0$, we obtain

$$(\mathbb{1}_A - \delta_l Z_A) \otimes (B_{l,1} B_{l,0} + B_{l,0} B_{l,1}) |\tilde{\psi}\rangle = 0. \quad (\text{C29})$$

Because $\alpha_i \neq 0$ for any i , we have $\delta_l \neq 1$ for any l . Therefore, $\mathbb{1}_A - \delta_l Z_A$ is invertible, which, using the fact that the

local states are full rank, allows us to conclude from the above formula that

$$\{B_{l,0}, B_{l,1}\} = 0. \quad (\text{C30})$$

Since the operators $B_{l,0}$ and $B_{l,1}$ anticommute, we can use the result from [13] to guarantee that there exist unitaries U_{B_l} for every $l = 1, \dots, N-1$ such that

$$U_{B_l} B_{l,0} U_{B_l}^\dagger = Z \otimes \mathbb{1}, \quad U_{B_l} B_{l,1} U_{B_l}^\dagger = X \otimes \mathbb{1}. \quad (\text{C31})$$

Let us now find the state that maximally violates the steering inequality provided by the operator (C2). As the Hilbert spaces of all parties decompose as $\mathbb{C}^2 \otimes H_{B_l'}$, we consider a state in a general form as

$$\bigotimes_{l=1}^{N-1} U_{B_l} |\psi\rangle = \sum_{i_0, i_1, \dots, i_{N-1}=0,1} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle, \quad (\text{C32})$$

where U_{B_l} are the unitaries in (C31) and $|\psi_{i_0 i_1 \dots i_{N-1}}\rangle$ are some unnormalized states. We can multiply (C13) by $\bigotimes_{l=1}^{N-1} U_{B_l}$ to obtain

$$Z_A \otimes \bigotimes_{l=1}^{N-1} (U_{B_l} B_{l,0} U_{B_l}^\dagger) U_{B_l} |\psi\rangle = - \bigotimes_{l=1}^{N-1} U_{B_l} |\psi\rangle. \quad (\text{C33})$$

Now, using Eq. (C31) and inserting the state from (C32), the above formula can be expressed as

$$\begin{aligned} & \sum_{i_0, i_1, \dots, i_{N-1}=0,1} (-1)^{1+i_0+i_1+\dots+i_{N-1}} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle \\ &= \sum_{i_0, i_1, \dots, i_{N-1}=0,1} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle. \end{aligned} \quad (\text{C34})$$

The above relation implies that the state (C32) is of the form

$$\begin{aligned} \bigotimes_{l=1}^{N-1} U_{B_l} |\psi\rangle &= \sum_{i_0, i_1, \dots, i_{N-1}=0,1} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle \text{ with} \\ &\times i_0 + i_1 + \dots + i_{N-1} = 2n + 1, \end{aligned} \quad (\text{C35})$$

for some non-negative integer n .

Now, we consider relation (C14) for any l and multiply it by $(\mathbb{1} - P_l)$ to obtain, for any l ,

$$Z_A \otimes \mathbb{1}_l \otimes (\mathbb{1} - P_l) |\psi\rangle = (\mathbb{1} - P_l) |\psi\rangle. \quad (\text{C36})$$

As $B_{j,0}$ are certified in (C31), we obtain $P_l = |0 \dots 0\rangle\langle 0 \dots 0| \otimes \mathbb{1}_{B_1' \dots B_{N-1}'}$. Now, plugging the state (C35) into the above

formula, we get

$$\begin{aligned} & \sum_{\substack{i_0, i_1, \dots, i_{N-1}=0,1 \\ i_1 + \dots + i_{N-1} - i_l \neq 0}} (-1)^{i_0} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle \\ &= \sum_{\substack{i_0, i_1, \dots, i_{N-1}=0,1 \\ i_1 + \dots + i_{N-1} - i_l \neq 0}} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle. \end{aligned} \quad (\text{C37})$$

This implies from (C35) that

$$|\psi_{i_1 \dots i_{N-1}}\rangle = 0 \quad \text{s.t.} \quad i_1 + \dots + i_{N-1} = 2n(n \neq 0). \quad (\text{C38})$$

Now, considering (C15) for any l and multiplying it by $\mathbb{1} - P_l$, we obtain

$$\mathbb{1}_A \otimes B_{l,0} \otimes (\mathbb{1} - P_l) |\psi\rangle = (\mathbb{1} - P_l) |\psi\rangle. \quad (\text{C39})$$

Then, plugging in the state (C35) with the certified observables (C31), we obtain

$$\begin{aligned} & \sum_{\substack{i_0, i_1, \dots, i_{N-1}=0,1 \\ i_1 + \dots + i_{N-1} - i_l \neq 0}} (-1)^{i_l} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle \\ &= \sum_{\substack{i_0, i_1, \dots, i_{N-1}=0,1 \\ i_1 + \dots + i_{N-1} - i_l \neq 0}} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle. \end{aligned} \quad (\text{C40})$$

This implies from (C35) that

$$\begin{aligned} |\psi_{i_0 i_1 \dots i_{N-1}}\rangle &= 0 \quad \text{s.t.} \\ i_0 + i_1 + \dots + i_{N-1} - i_l &= 2n(n \neq 0). \end{aligned} \quad (\text{C41})$$

It is now straightforward to observe from Eqs. (C35), (C38), and (C41) that the state $|\psi\rangle$ that attains the quantum bound of $\langle \mathcal{W}_N \rangle$ is of the form

$$\begin{aligned} \bigotimes_{l=1}^{N-1} U_{B_l} |\psi\rangle &= \sum_{i_0, i_1, \dots, i_{N-1}=0,1} |i_0 i_1 \dots i_{N-1}\rangle |\psi_{i_0 i_1 \dots i_{N-1}}\rangle \text{ with} \\ i_0 + i_1 + \dots + i_{N-1} &= 1. \end{aligned} \quad (\text{C42})$$

Let us now consider (C14) and multiply it by P_l to obtain, for any l ,

$$(\gamma_l X_A \otimes B_{l,1} + \delta_l Z_A \otimes \mathbb{1}_l) \otimes P_l |\psi\rangle = P_l |\psi\rangle. \quad (\text{C43})$$

Plugging in the state from (C42) and the observables from (C31), we obtain the following two conditions:

$$\begin{aligned} \gamma_l |\psi_{0 \dots 1_l \dots 0_{N-1}}\rangle - \delta_l |\psi_{1 \dots 0_l \dots 0_{N-1}}\rangle &= |\psi_{1 \dots 0_l \dots 0_{N-1}}\rangle, \\ \gamma_l |\psi_{1 \dots 0_l \dots 0_{N-1}}\rangle + \delta_l |\psi_{0 \dots 1_l \dots 0_{N-1}}\rangle &= |\psi_{0 \dots 1_l \dots 0_{N-1}}\rangle. \end{aligned}$$

Using the values of γ_l and δ_l from (C3), the solution of these two relations is $|\psi_{0 \dots 1_l \dots 0_{N-1}}\rangle = (\alpha_{l+1})/\alpha_1$

$|\psi_{1\dots 0_{l\dots 0_{N-1}}}\rangle$ for all $l = 1, \dots, N-1$. Therefore, the state that maximally violates our steering inequalities has the following form:

$$\begin{aligned} & \left(\mathbb{1}_{AE} \otimes \bigotimes_{l=1}^{N-1} U_{B_l} \right) |\psi\rangle_{ABE} \\ &= \left(\sum_{l=0}^{N-1} \alpha_{l+1} |0\dots, 0, 1_l, 0, \dots, 0\rangle \right) \otimes \left(\frac{1}{\alpha_1} |\psi_{1,0,\dots,0}\rangle \right), \end{aligned} \quad (\text{C44})$$

where $|\xi\rangle_{B''E} = 1/\alpha_1 |\psi_{1,0,\dots,0}\rangle$. ■

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